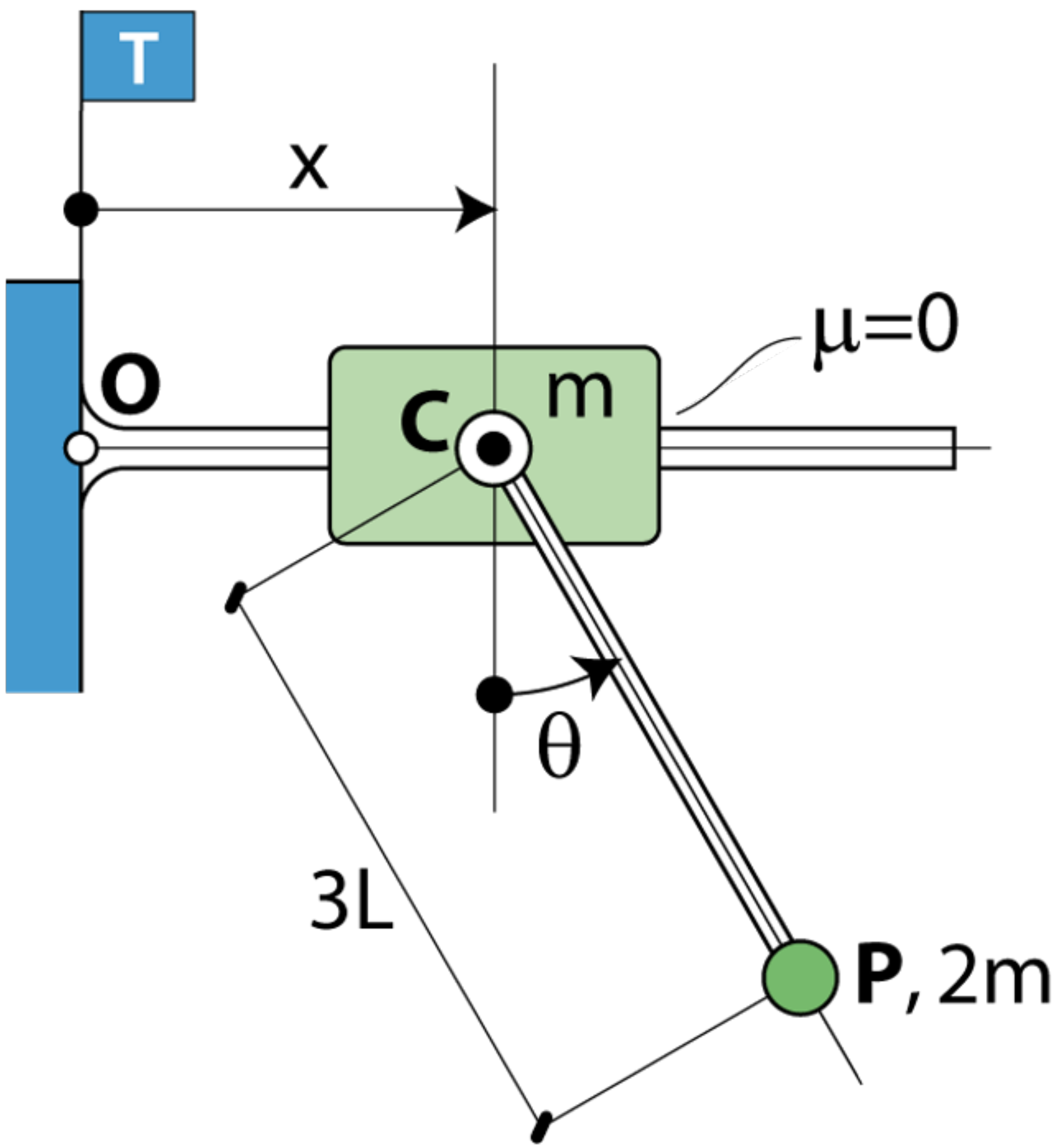
11P

Teoremes vectorials II

Exemples 3D



- Eqs. del moviment
  (sense cap motor instal·lat)

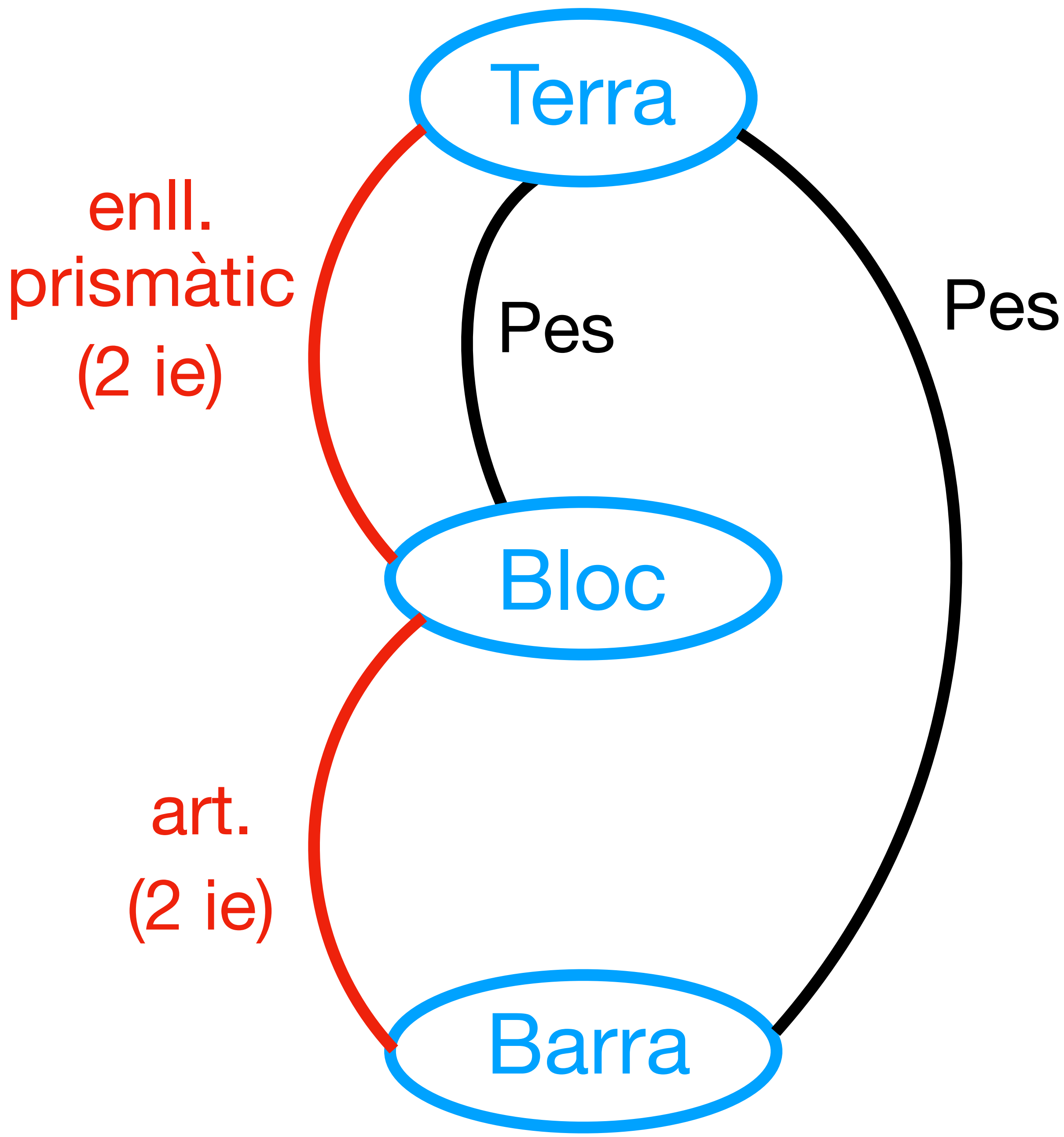
- x(t) si es posa un motor entre bloc i pèndol que garanteix $\dot{\theta} = \omega_0 = \text{ct}$

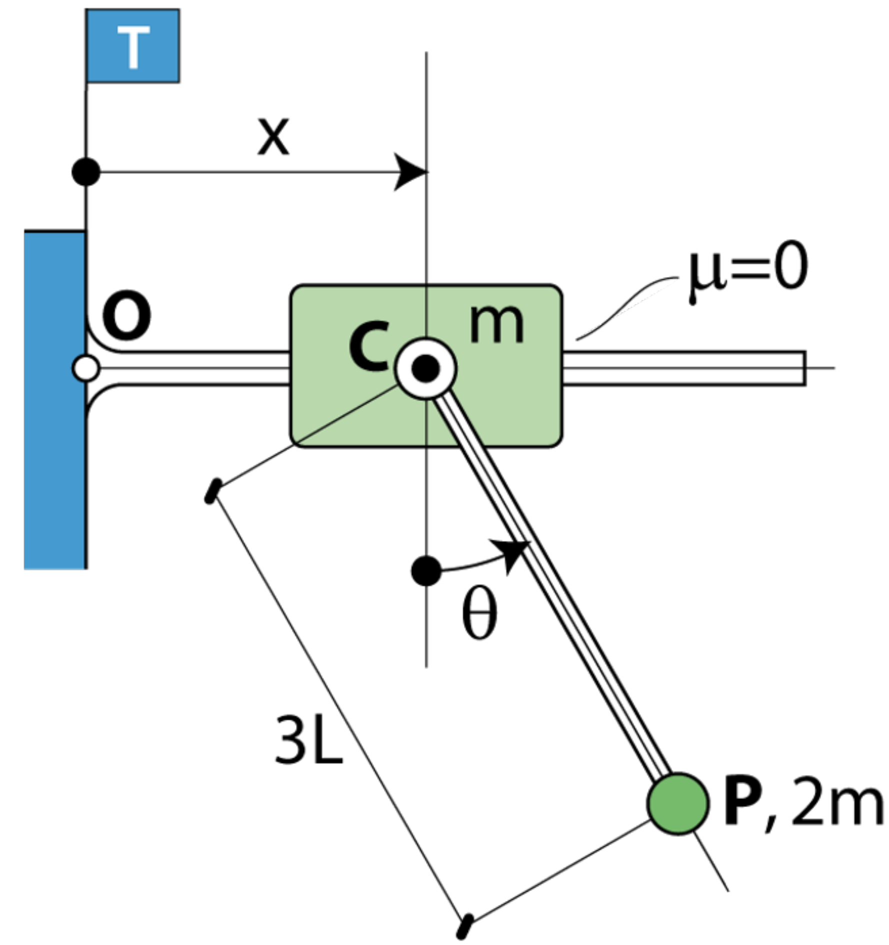
Condicions inicials
$$x(0) = 0$$
$$\dot{x}(0) = 0$$
$$\theta(0) = 0$$

Tractat com de dinàmica 2D

DGI

(sense motor)





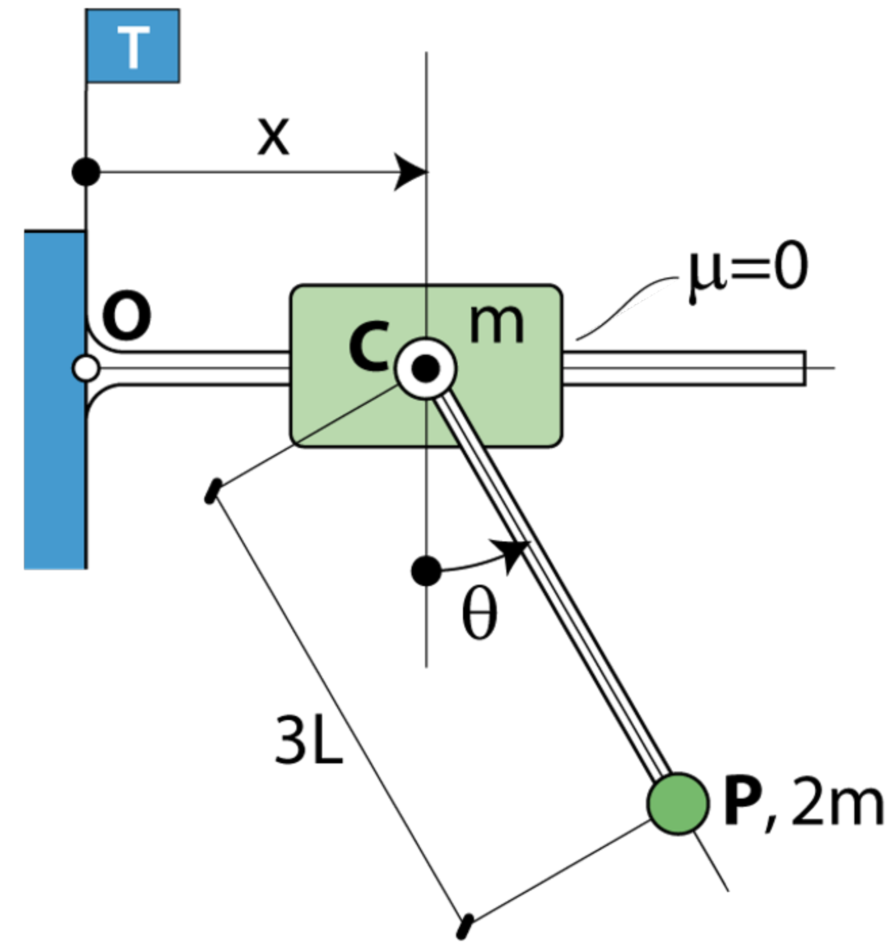
Problema global determinat

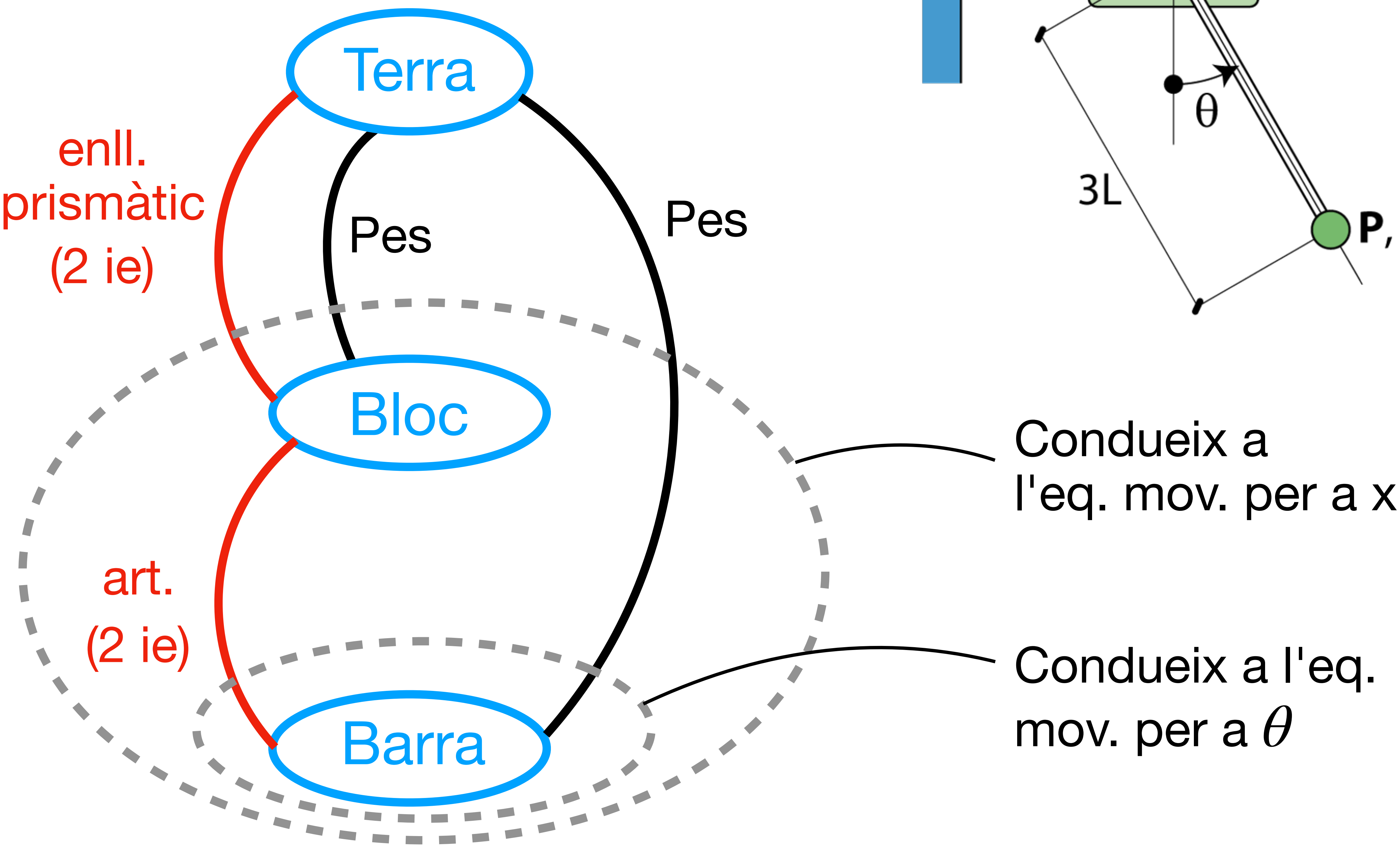
6 eqs = 2 sòlids · 3 eqs/sòlid

6 incògn. | 4 i.e.

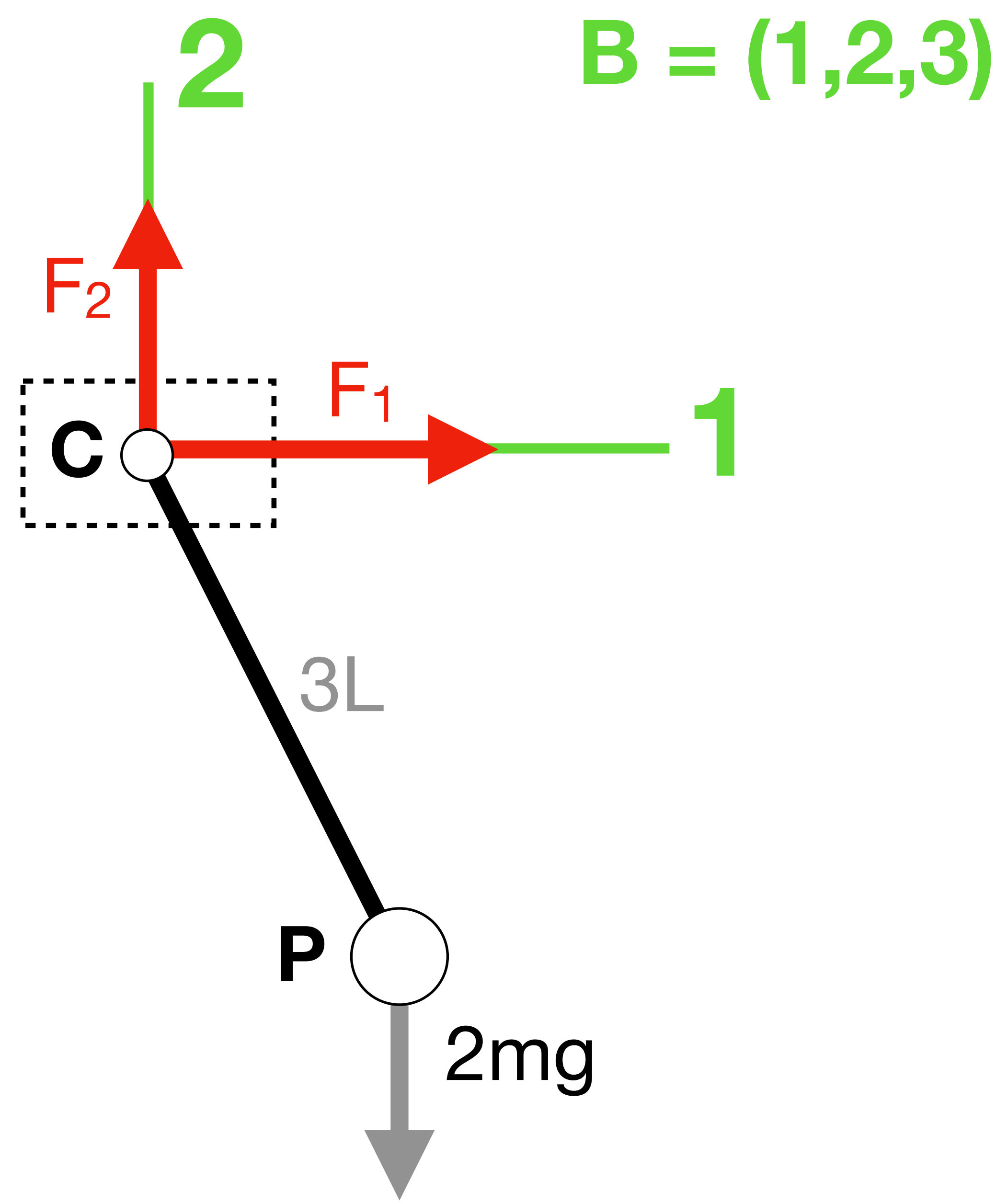
$\underbrace{\ddot{x}, \ddot{\theta}}_{\text{Associades a GL}}$

DGI
(sense motor)





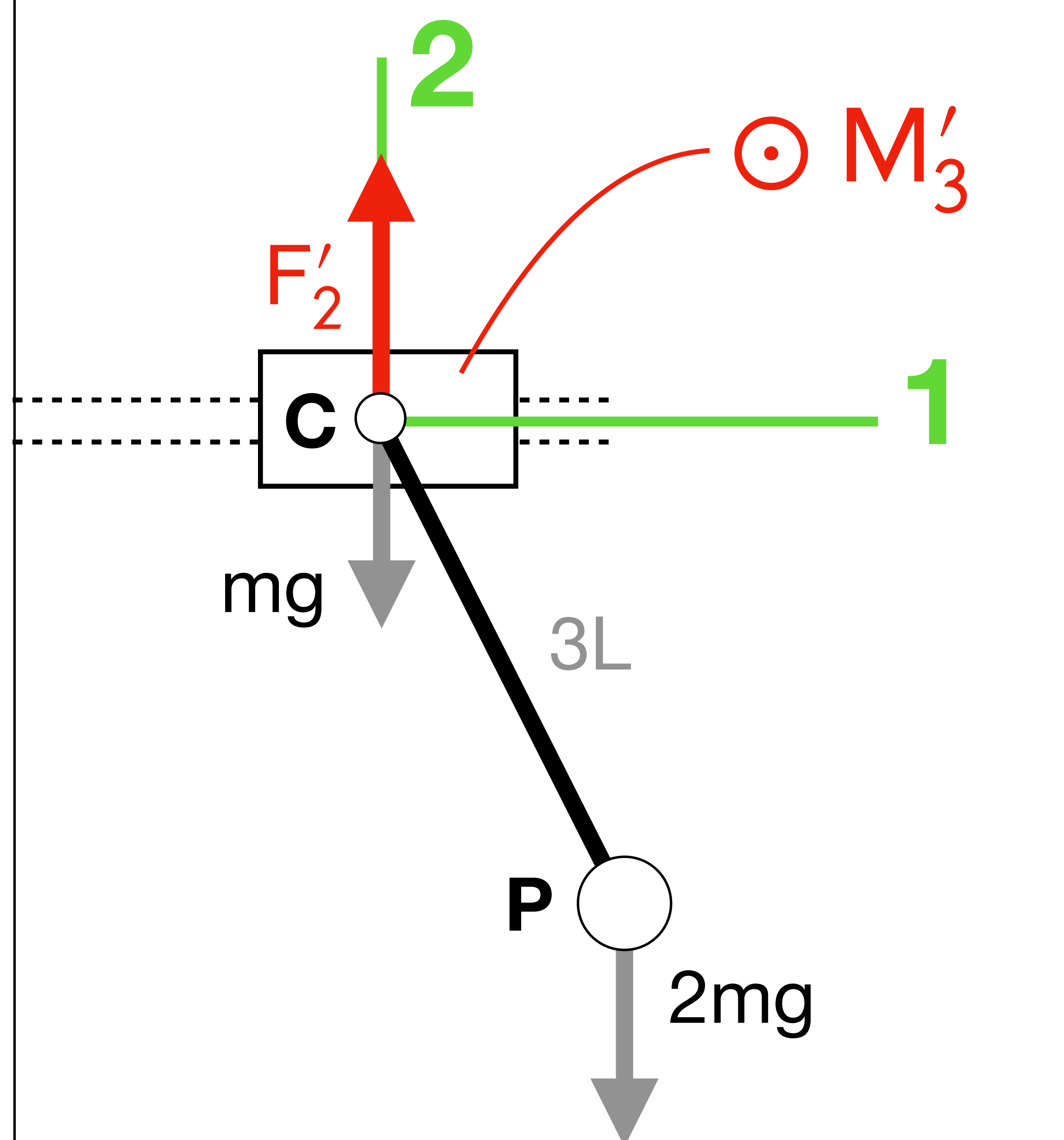
Sist = Barra



$$\left\{\bar{F}_{\text{Bloc}\rightarrow\text{Barra}}\right\}_B = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix}$$

$$\left\{\bar{M}_{\text{Bloc}\rightarrow\text{Barra}}(\mathbf{C})\right\}_B = \{0\}$$

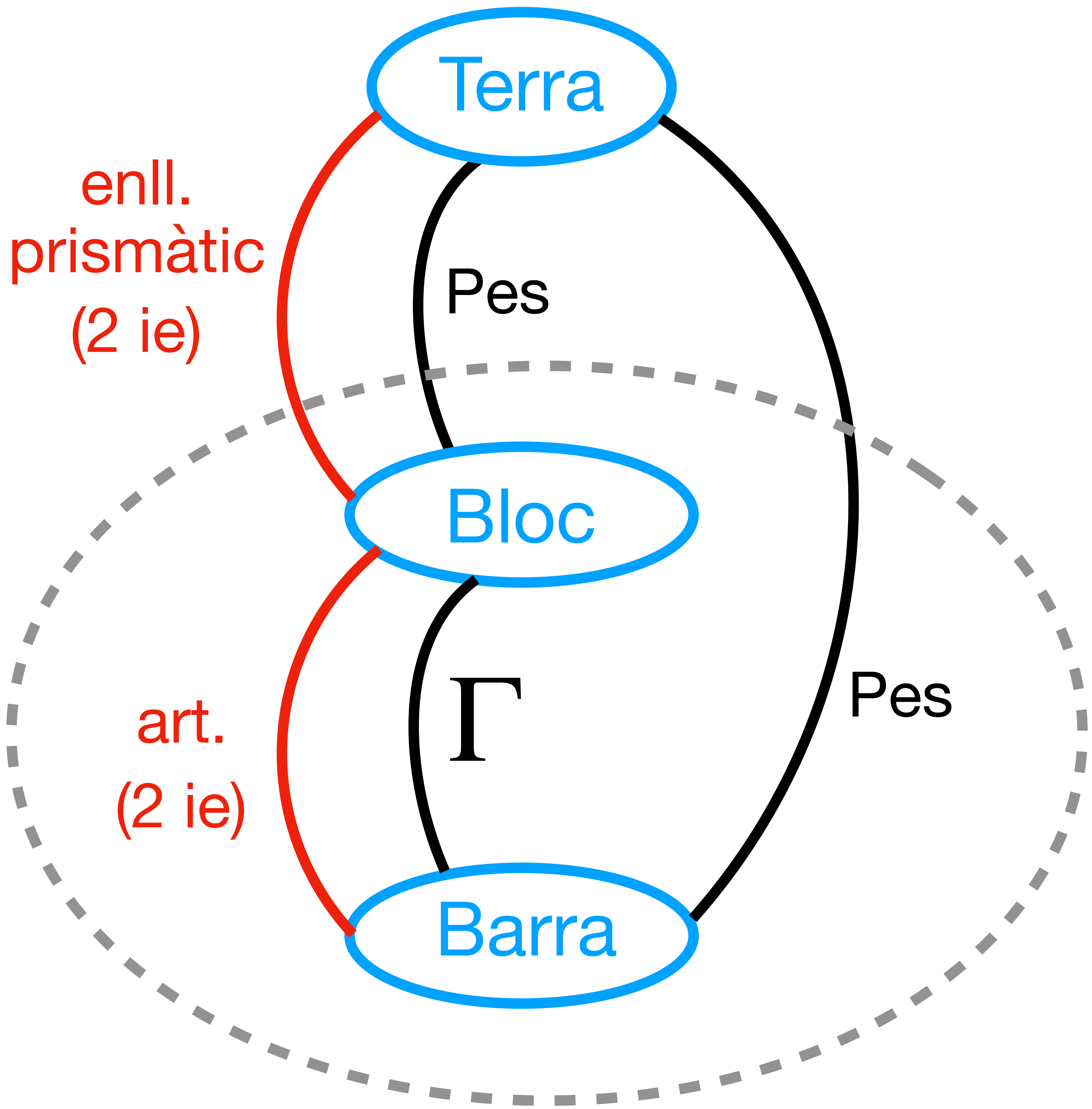
Sist = Bloc + Barra



$$\left\{\bar{F}_{T\rightarrow\text{Bloc}}\right\}_B = \begin{Bmatrix} 0 \\ F'_2 \end{Bmatrix}$$

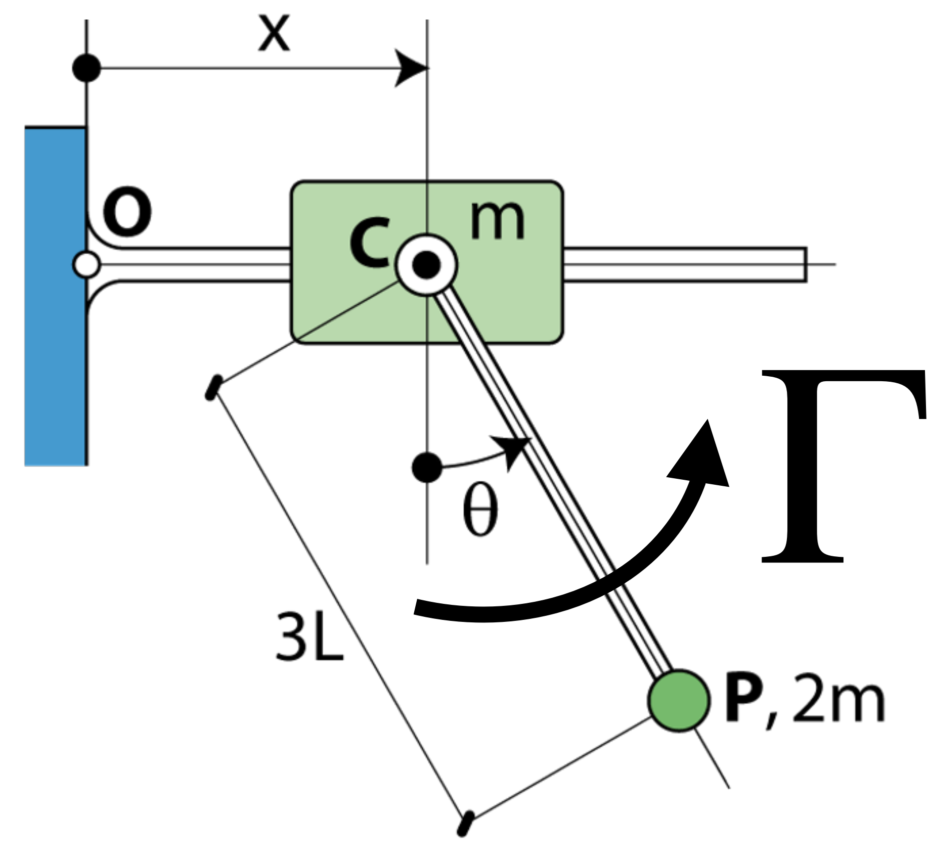
$$\left\{\bar{M}_{T\rightarrow\text{Bloc}}(\mathbf{C})\right\}_B = \{M'_3\}$$

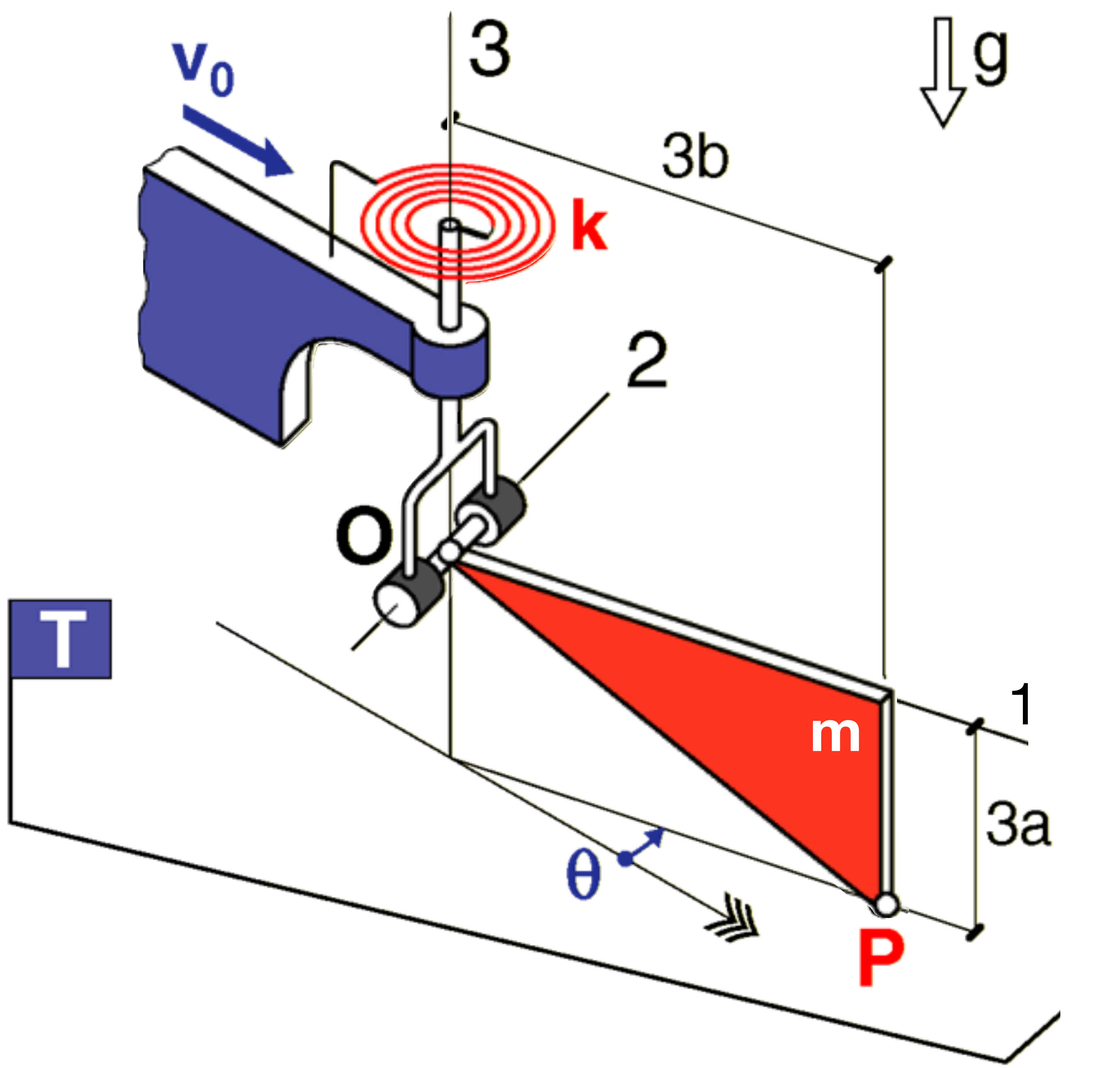
DGI
(amb motor)



Ara

Hi ha el parell estàtor→rotor



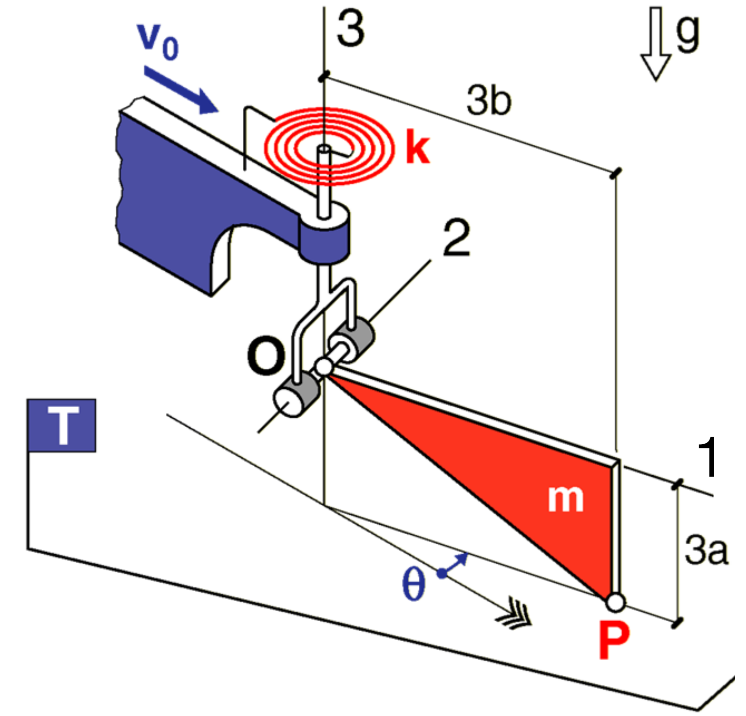


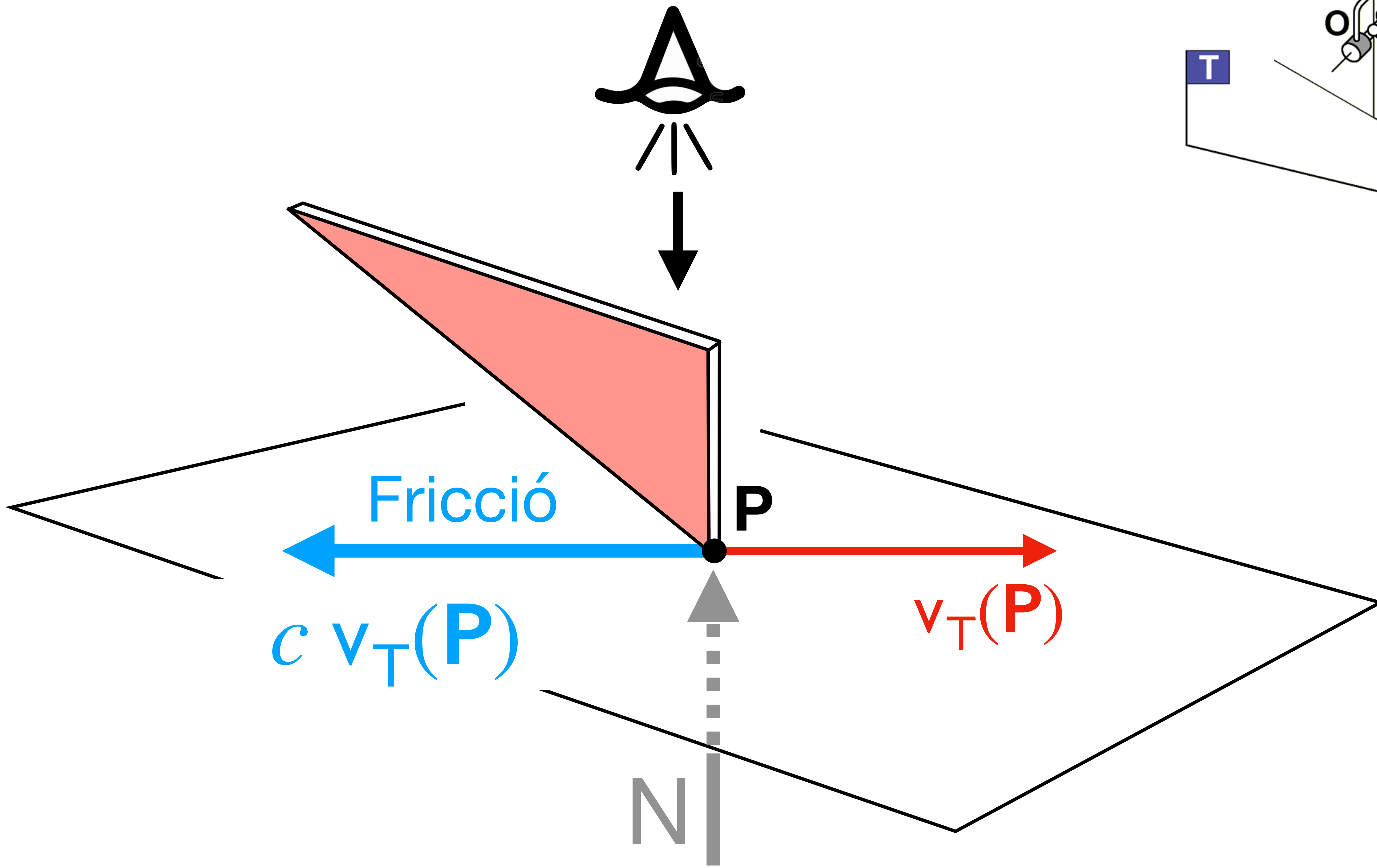
- Eq. mov. per a θ ?
- k_{min} per a que $\theta_{eq} = 0$ sigui **ESTABLE** ?

∃ frec viscós T → **P**
(de coef c)

Per $\theta = 0$ la molla **està distesa**

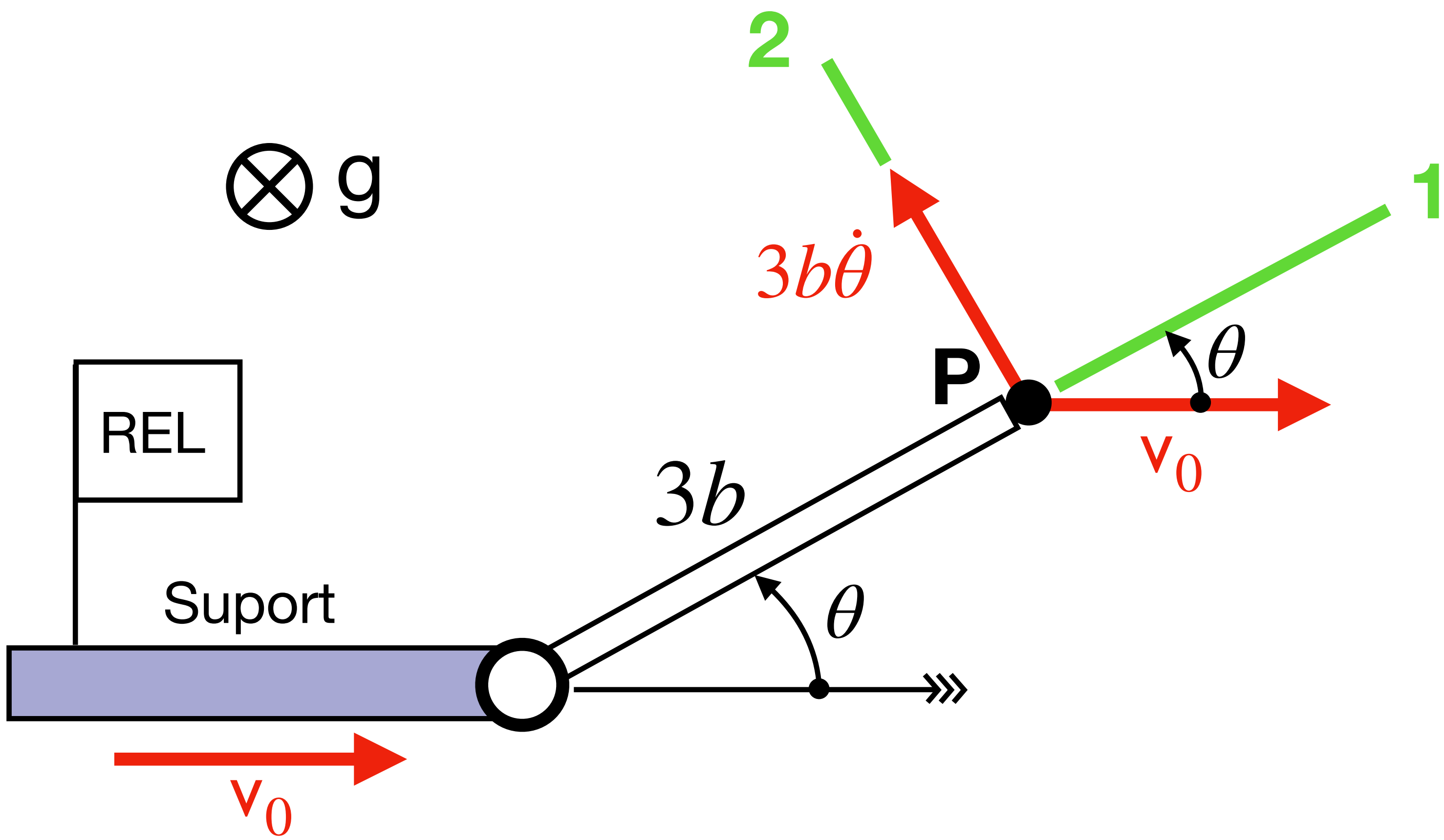
Força de frec viscós T → P

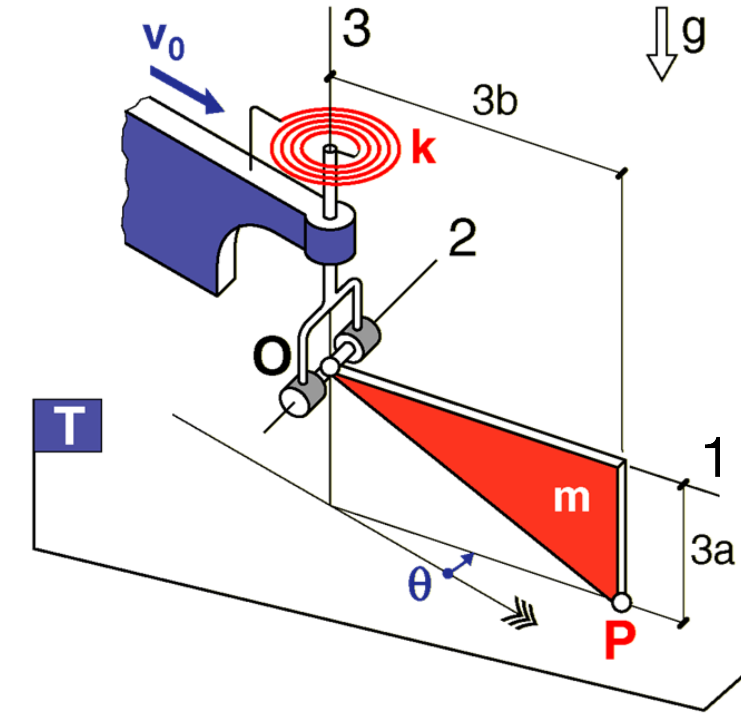




$$\boxed{\overline{\mathbf{F}}_{fv} = -\,c\,\overline{\mathrm{v}}_T(\mathbf{P})}$$

Força de frec viscós T → P

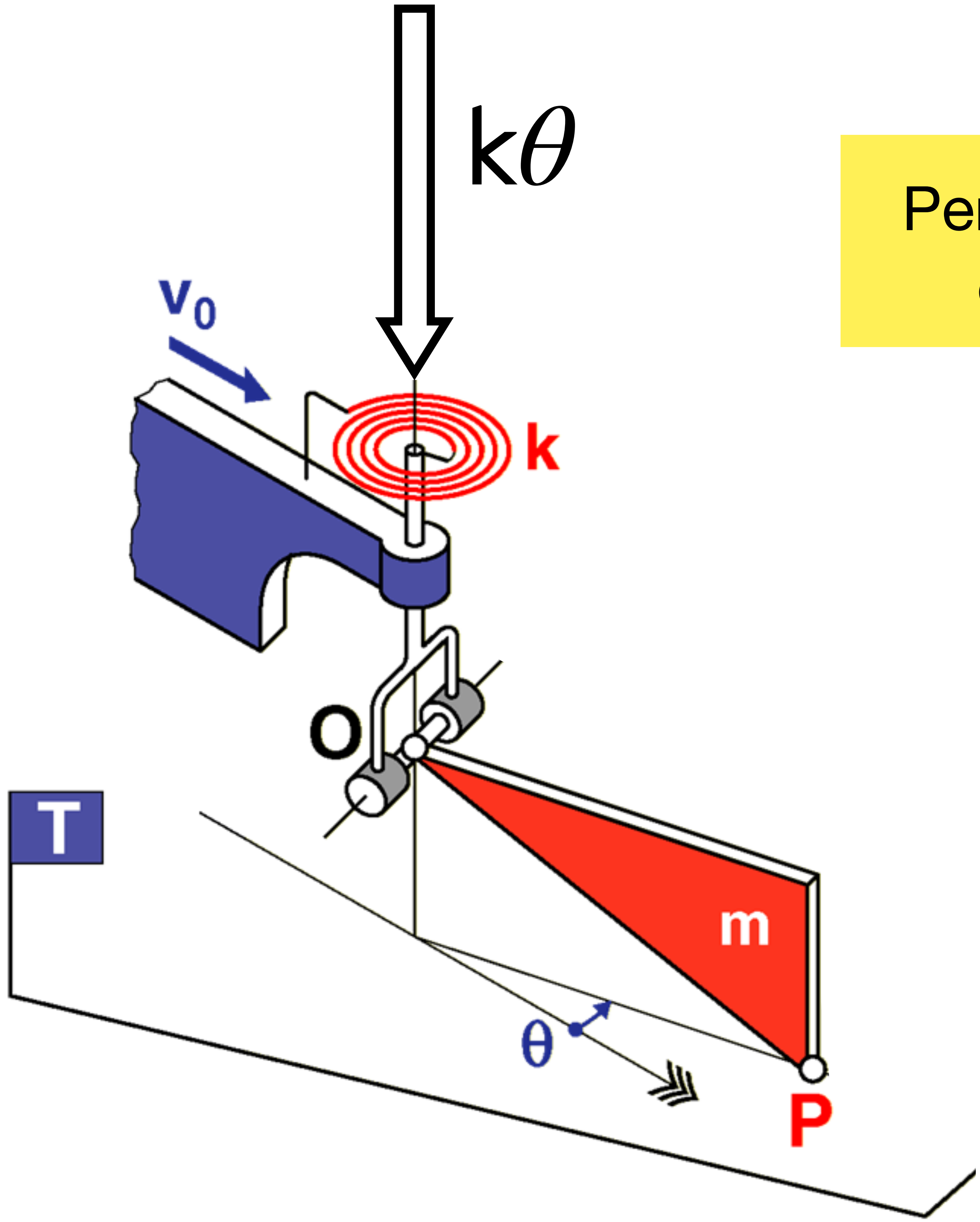




$$\bar{v}_T(\mathbf{P}) = \bar{v}_{REL}(\mathbf{P}) + \bar{v}_{ar}(\mathbf{P}) = \begin{Bmatrix} v_0\cos\theta \\ -v_0\sin\theta + 3b\dot{\theta} \\ 0 \end{Bmatrix}_{B=(1,2,3)}$$

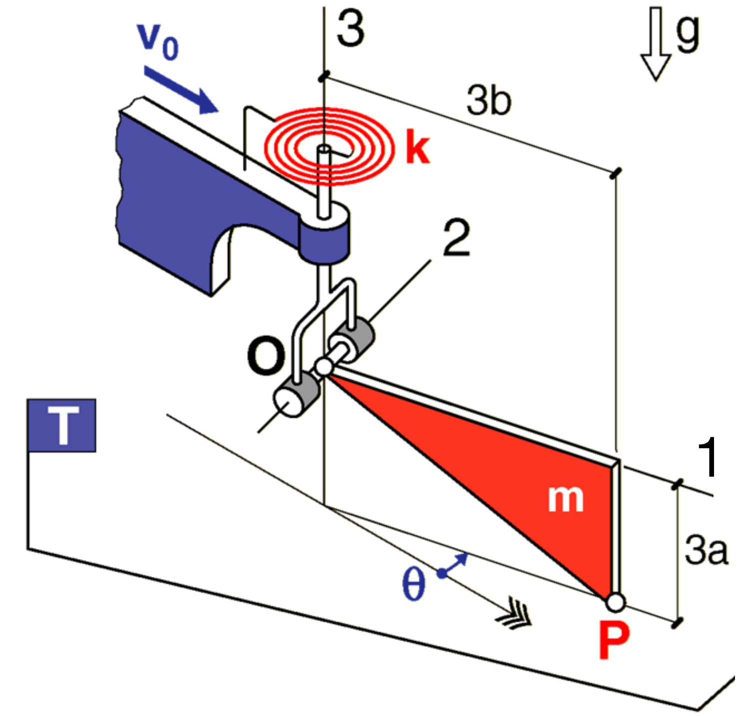
$$\bar{\mathbf{F}}_{fv} = -\,c\,\bar{v}_T(\mathbf{P}) = \begin{Bmatrix} \boxed{-cv_0\cos\theta} \\ \boxed{cv_0\sin\theta - 3cb\dot{\theta}} \\ 0 \end{Bmatrix}_B \quad \begin{matrix} F_{fv1} \\ F_{fv2} \\ \end{matrix}$$

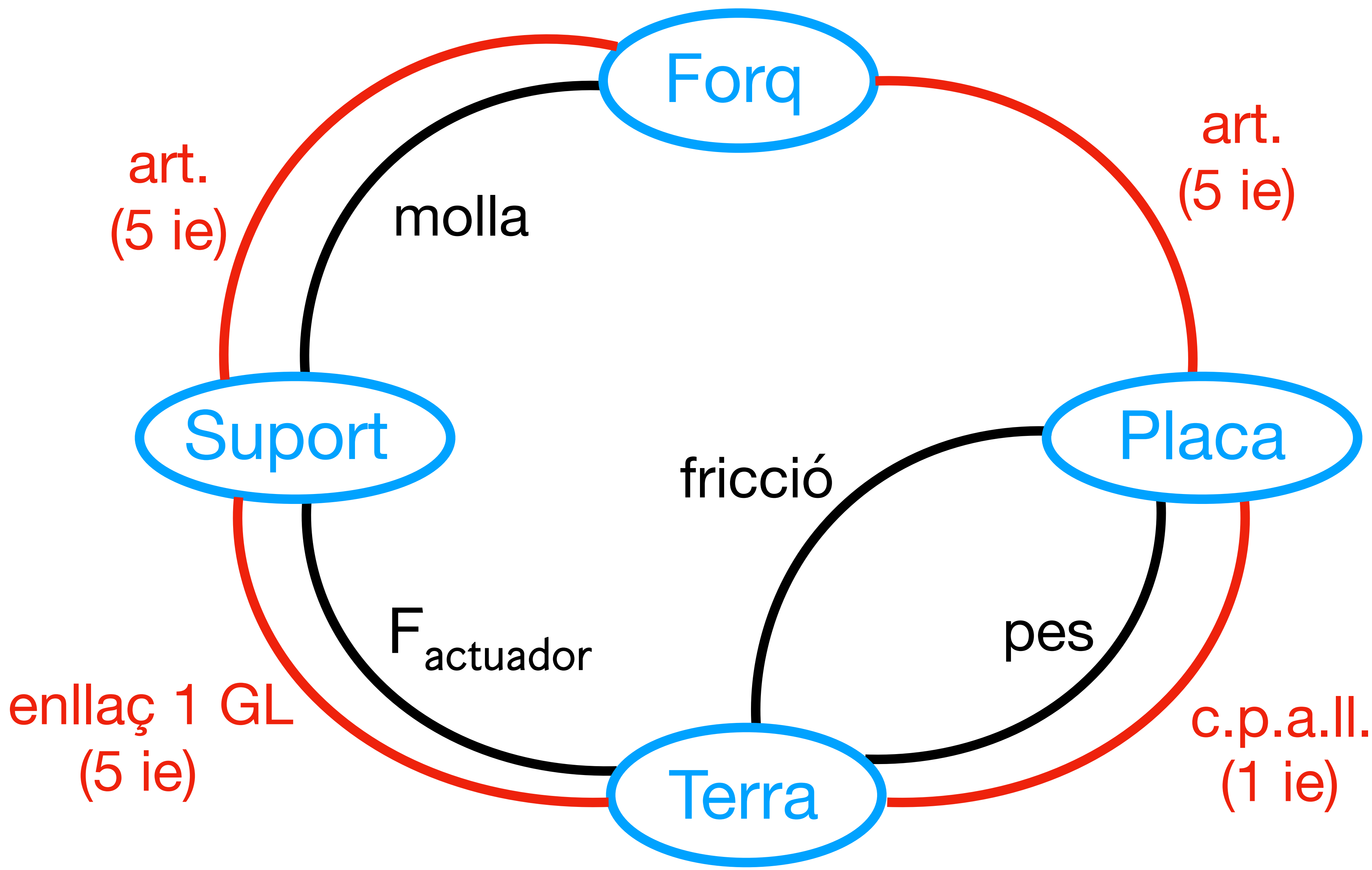
Parell molla torsional → forq



Per $\theta = 0$ la molla **està distesa**

DGI = Diagrama general d'interaccions

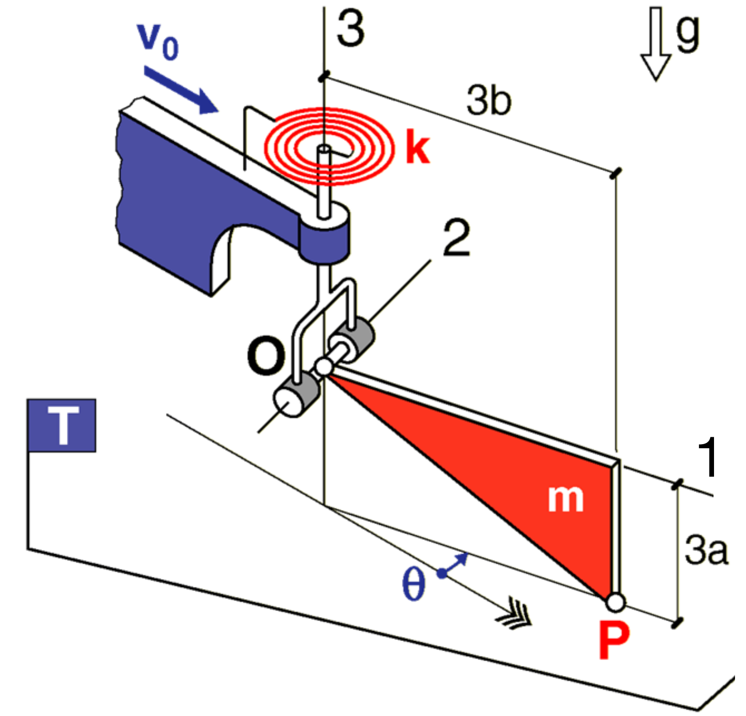


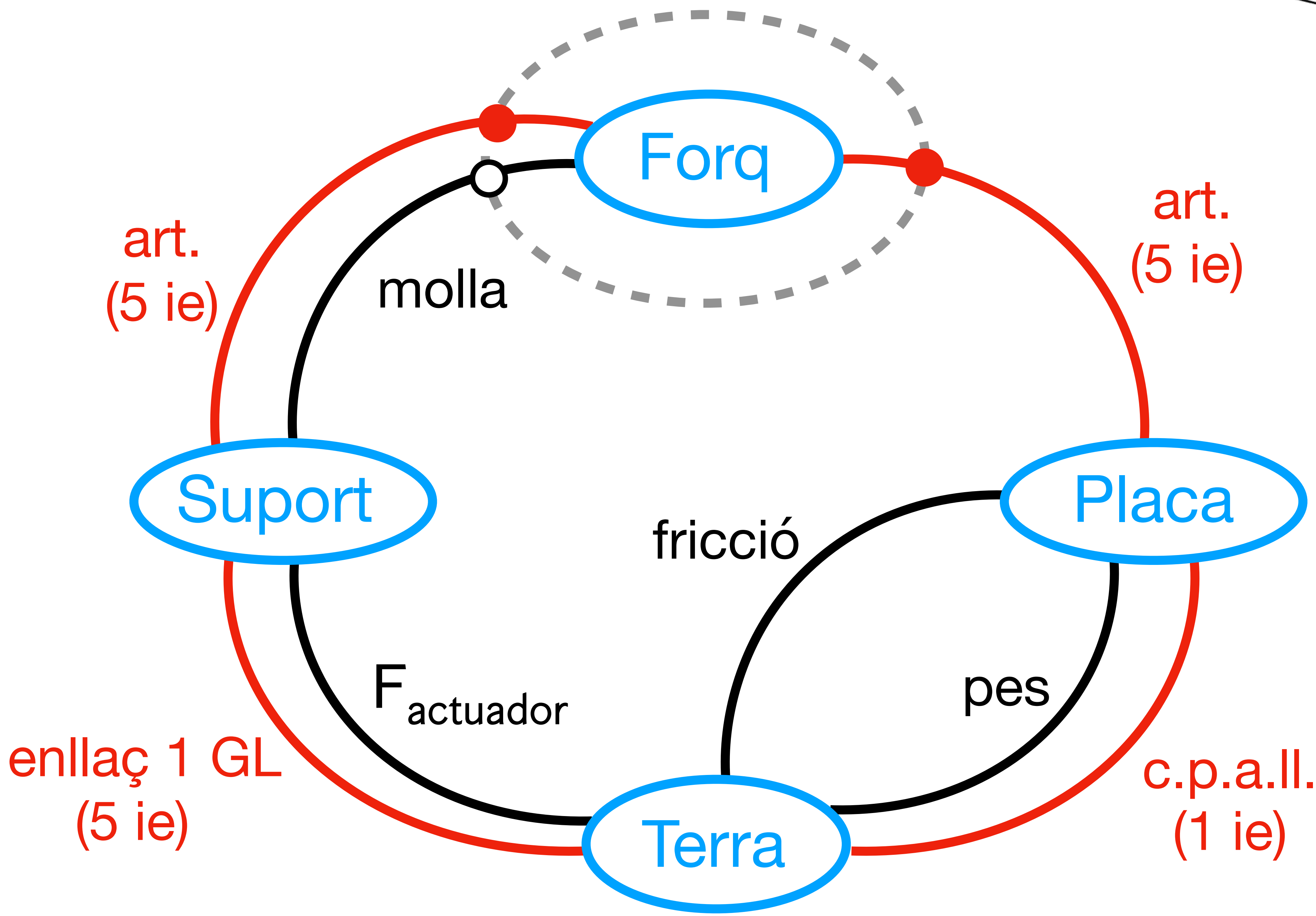


INDET

Sist = Forq

10 ie + $\ddot{\theta}$ = **11 incòg**

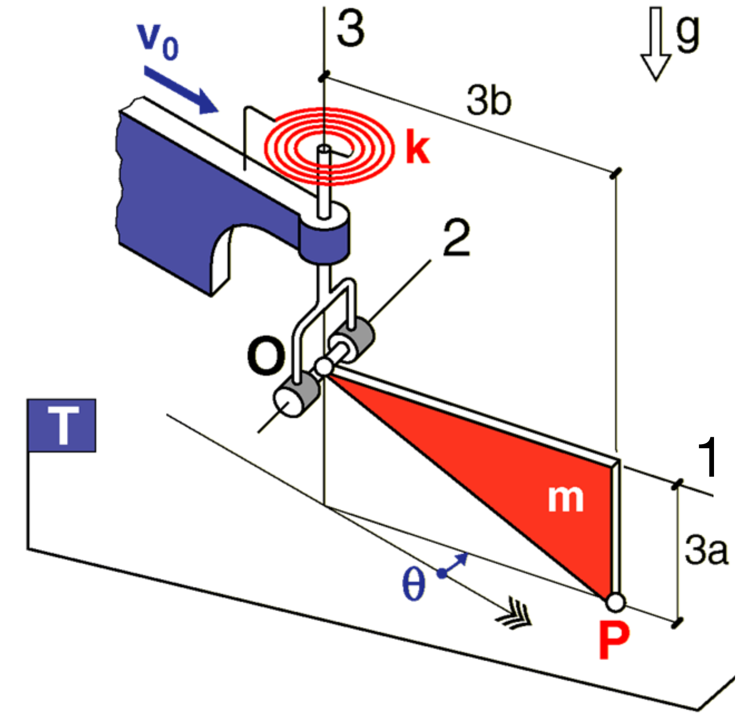


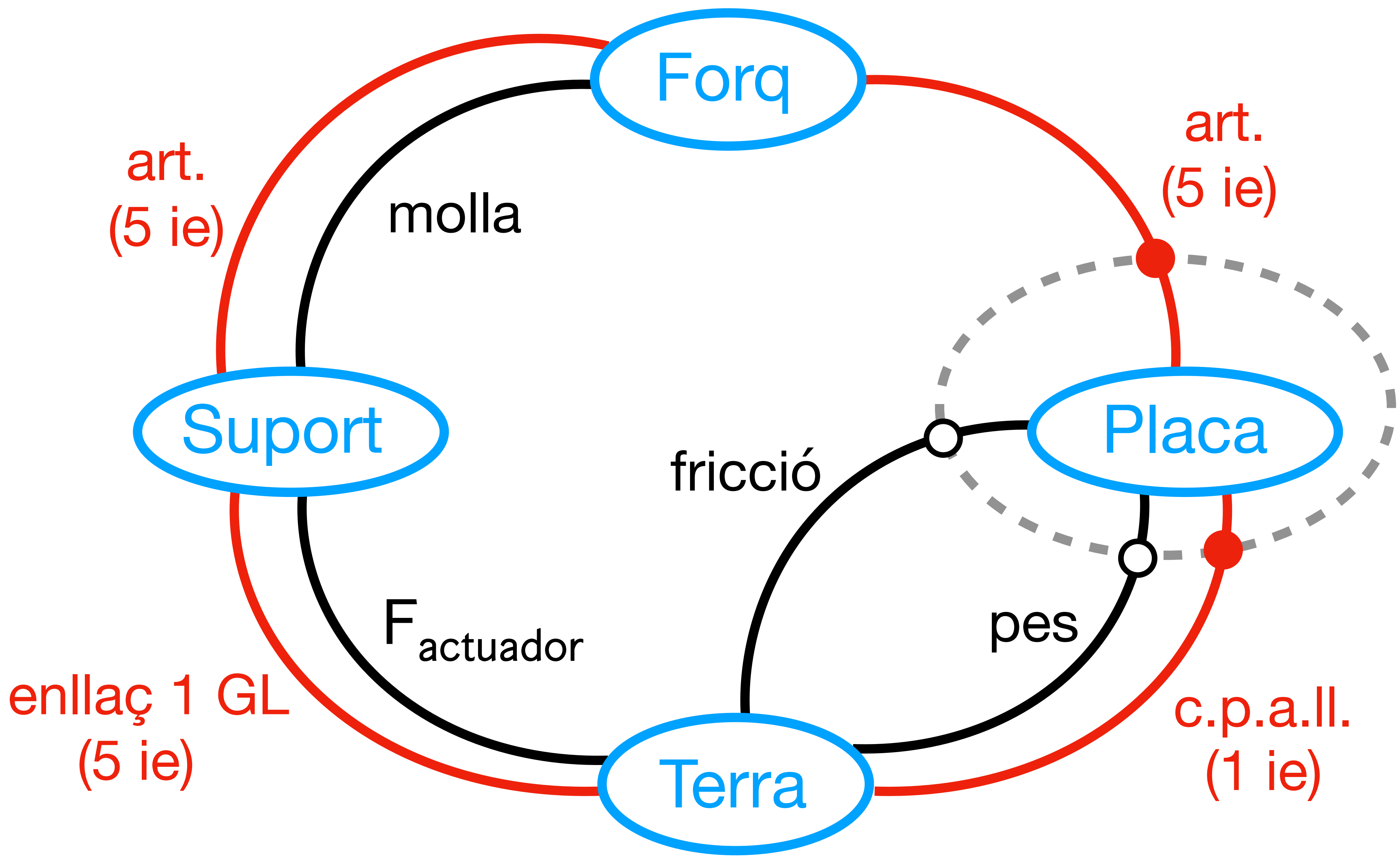


INDET

Sist = Placa

6 ie + $\ddot{\theta}$ = 7 incòg

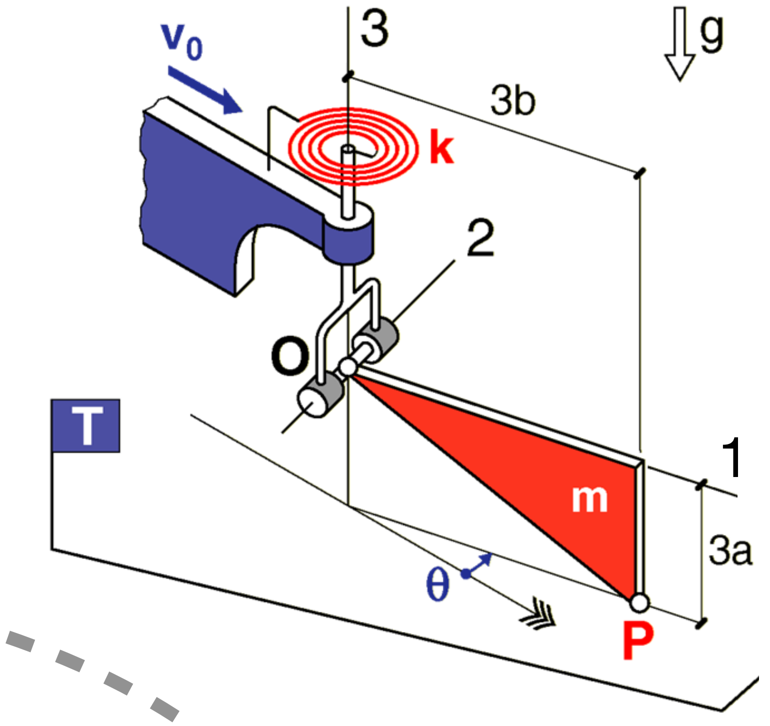


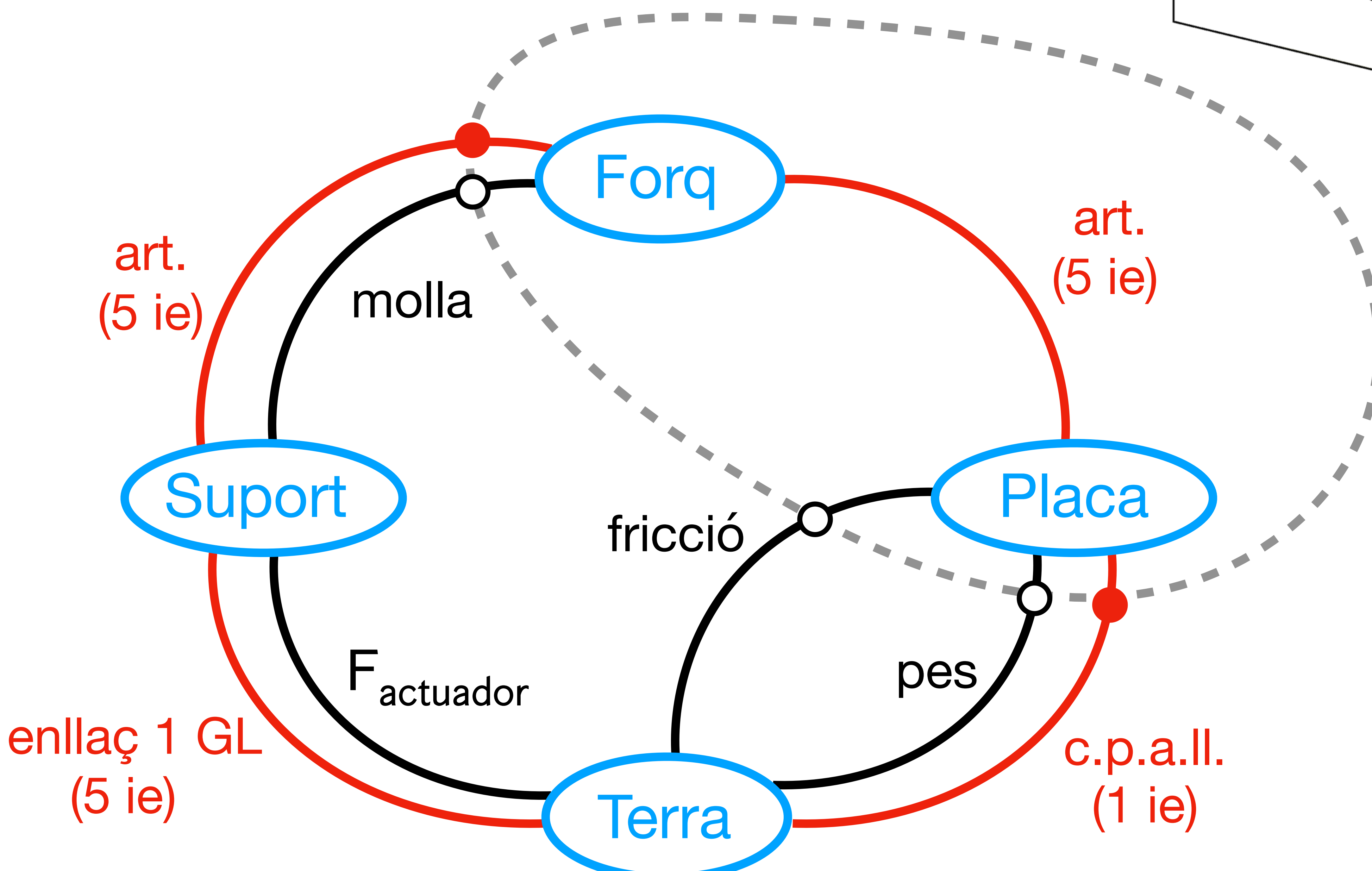


INDET

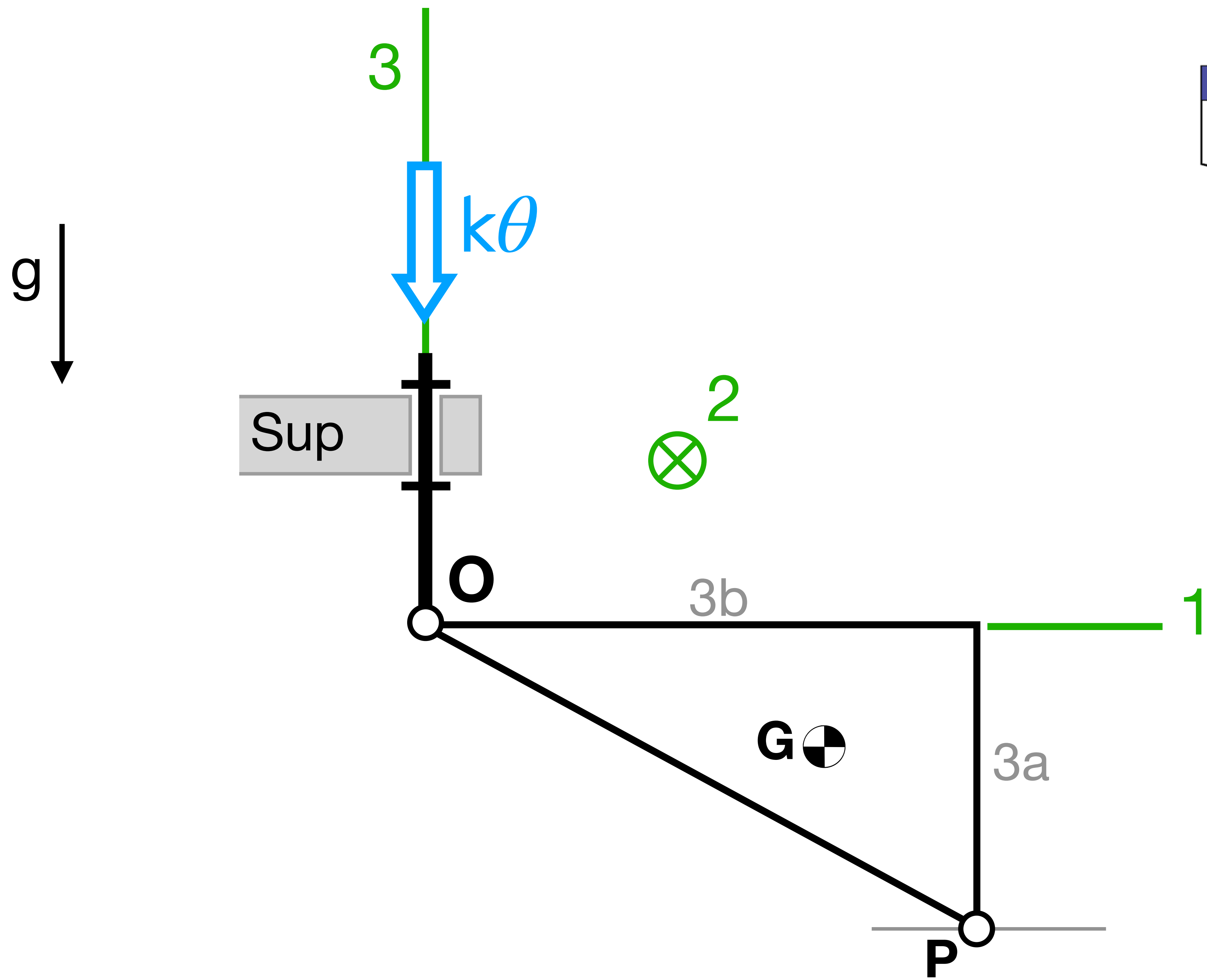
$$\text{Sist} = \text{Placa} + \text{Forq}$$

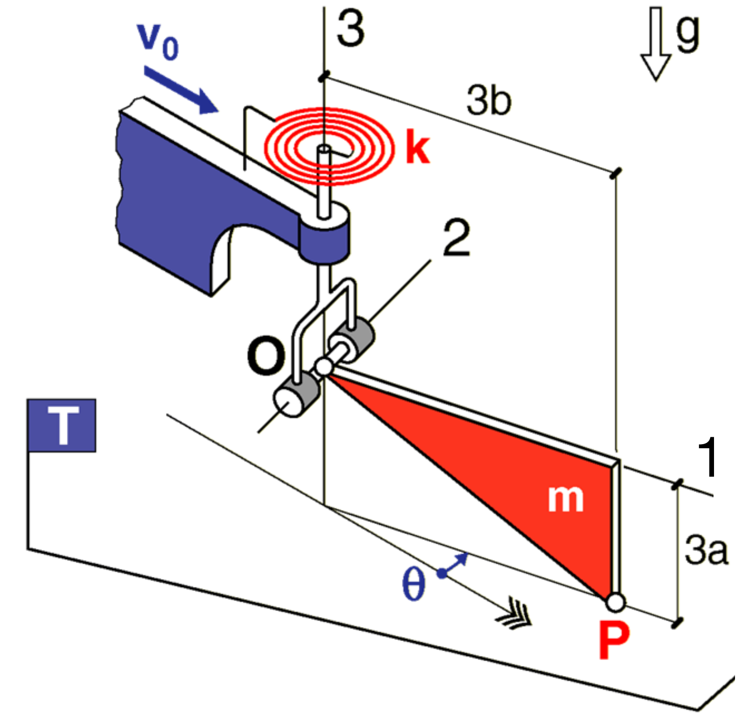
$$6 \text{ ie} + \ddot{\theta} = 7 \text{ incòg}$$



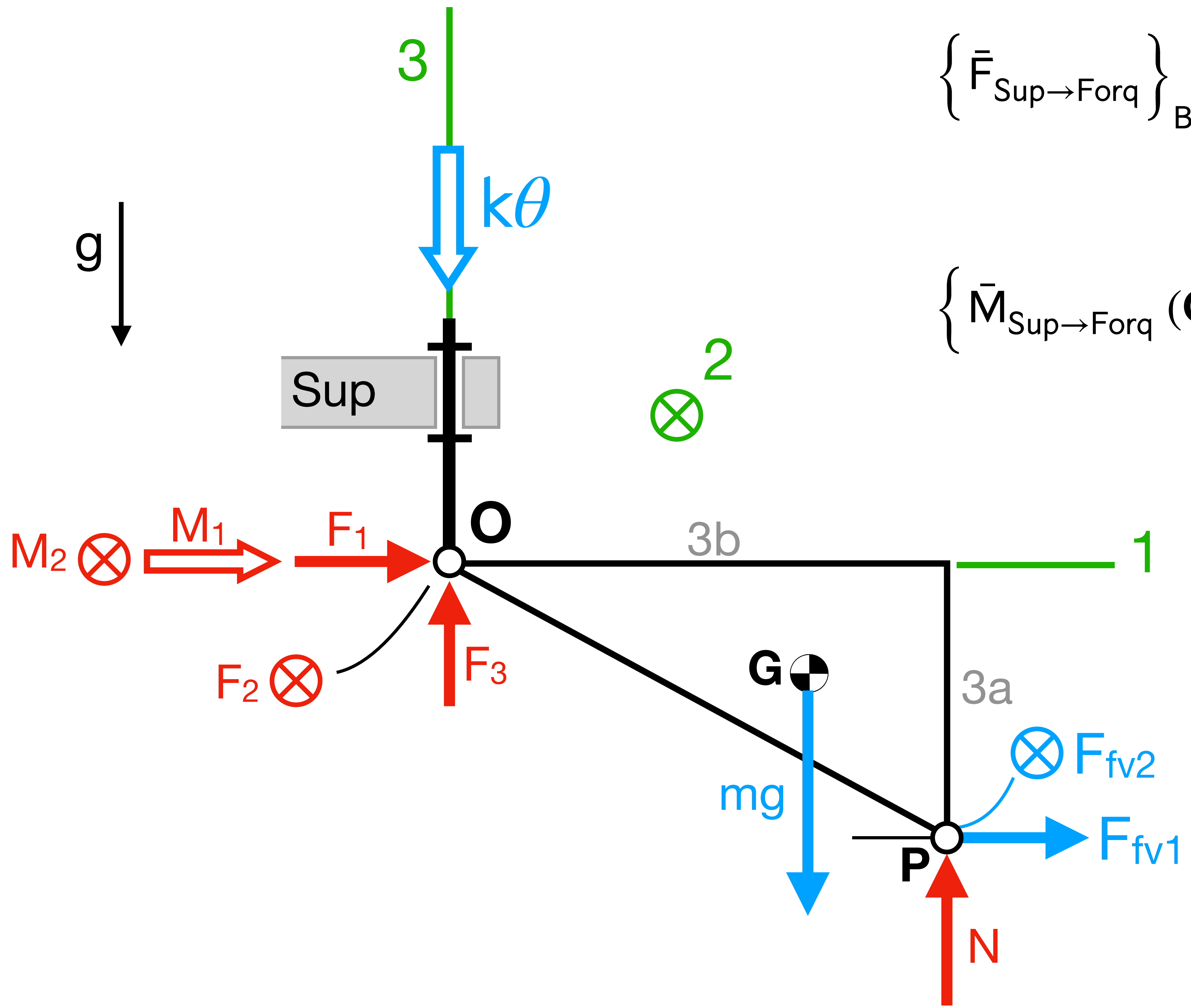


Interaccions sobre "Placa + Forq"





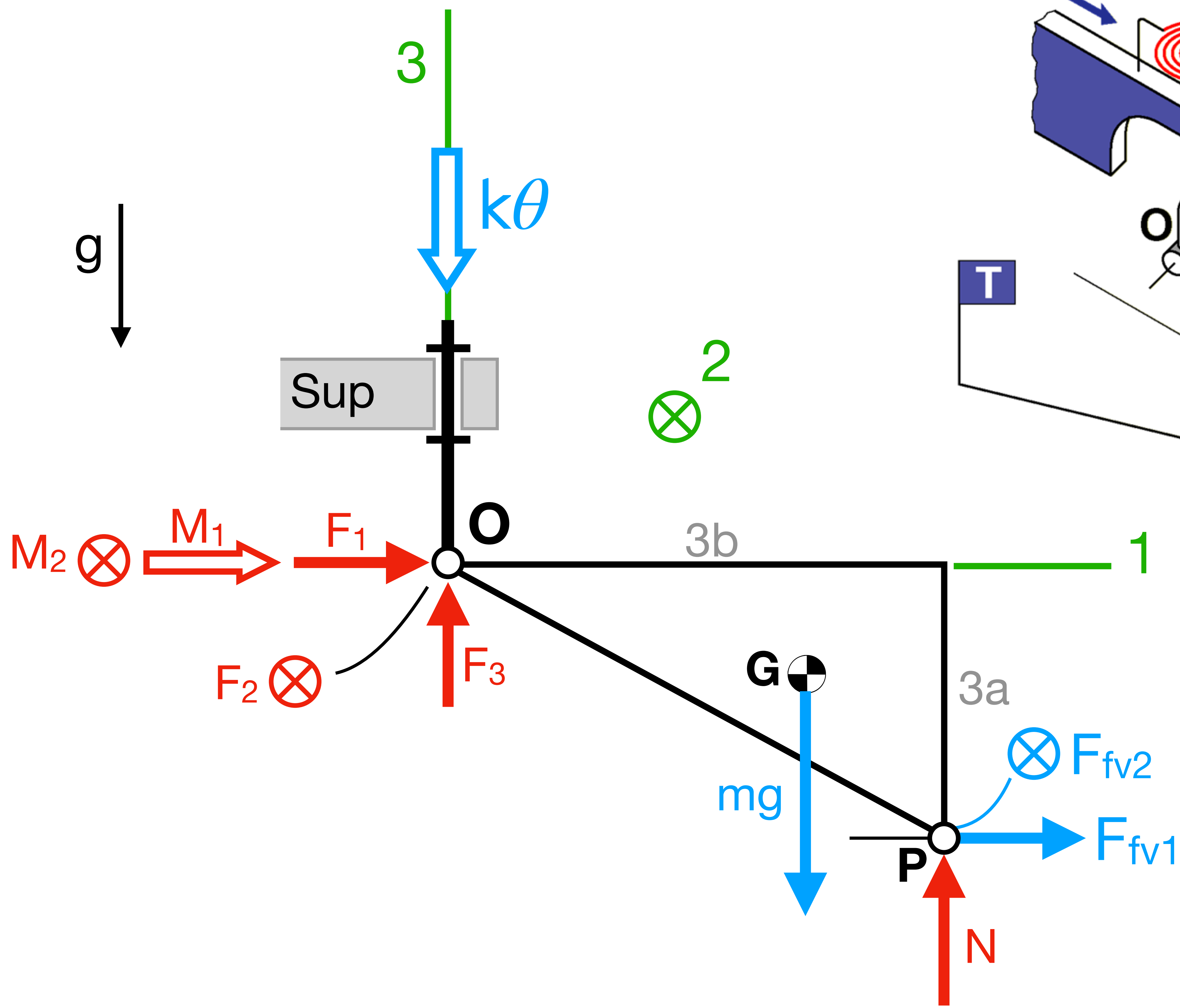
Interaccions sobre "Placa + Forq"

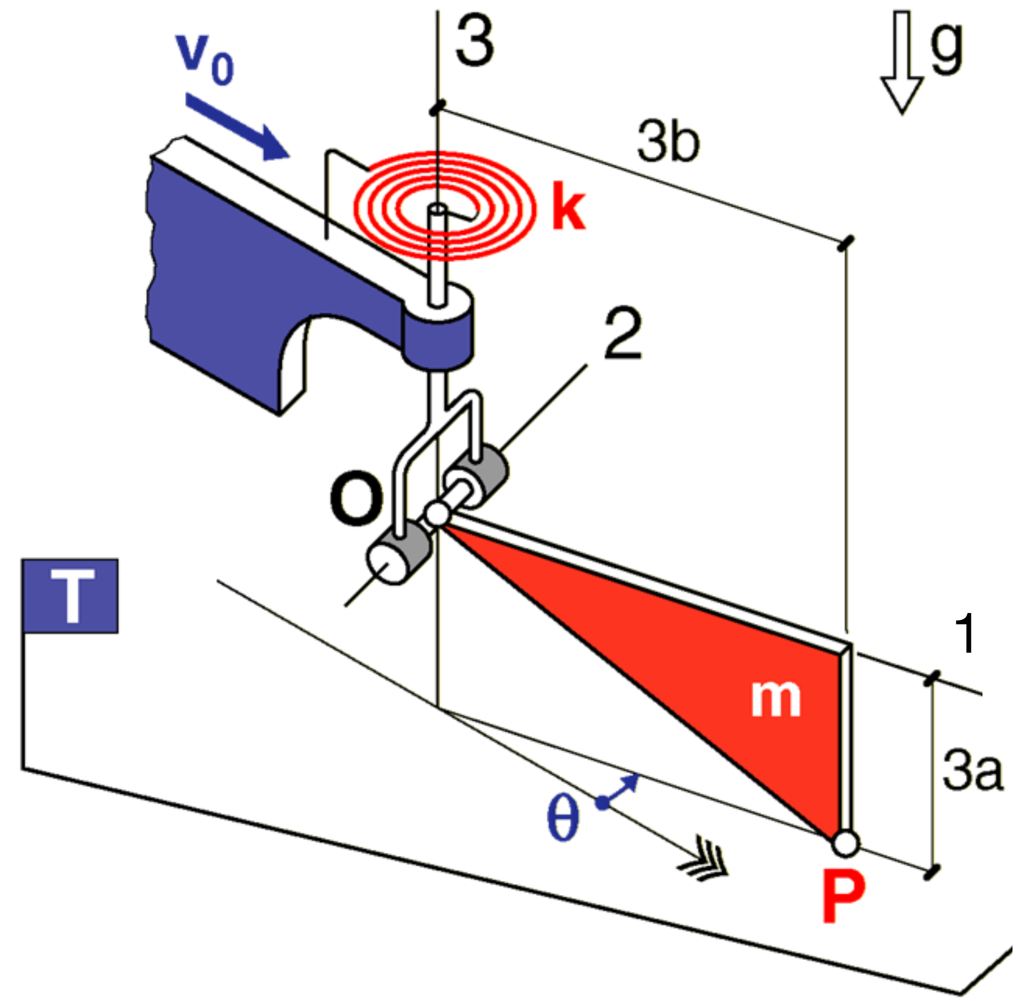


$$\left\{ \bar{F}_{Sup\rightarrow Forq} \right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\left\{ \bar{M}_{Sup\rightarrow Forq}\,(\mathbf{O}) \right\}_B = \begin{Bmatrix} M_1 \\ M_2 \\ 0 \end{Bmatrix}$$

Interaccions sobre "Placa + Forq"





Anàlisi de l'estabilitat de $\theta_{eq} = 0$

3 passos

com al pèndol simple

$$I_{33}\,\ddot{\theta} + 9cb^2\,\dot{\theta} + k\,\theta - 3bc\mathsf{v}_0\,\sin\theta = 0$$

Obtenim EDO de l'error ε

$$\theta = \theta_{\text{eq}} + \varepsilon = \varepsilon$$

en aquest exercici

$$\dot{\theta} = \dot{\varepsilon}$$

$$\ddot{\theta} = \ddot{\varepsilon}$$

$$I_{33}\,\ddot{\varepsilon} + 9cb^2\,\dot{\varepsilon} + k\,\varepsilon - 3bc\mathsf{v}_0\,\sin\varepsilon = 0$$

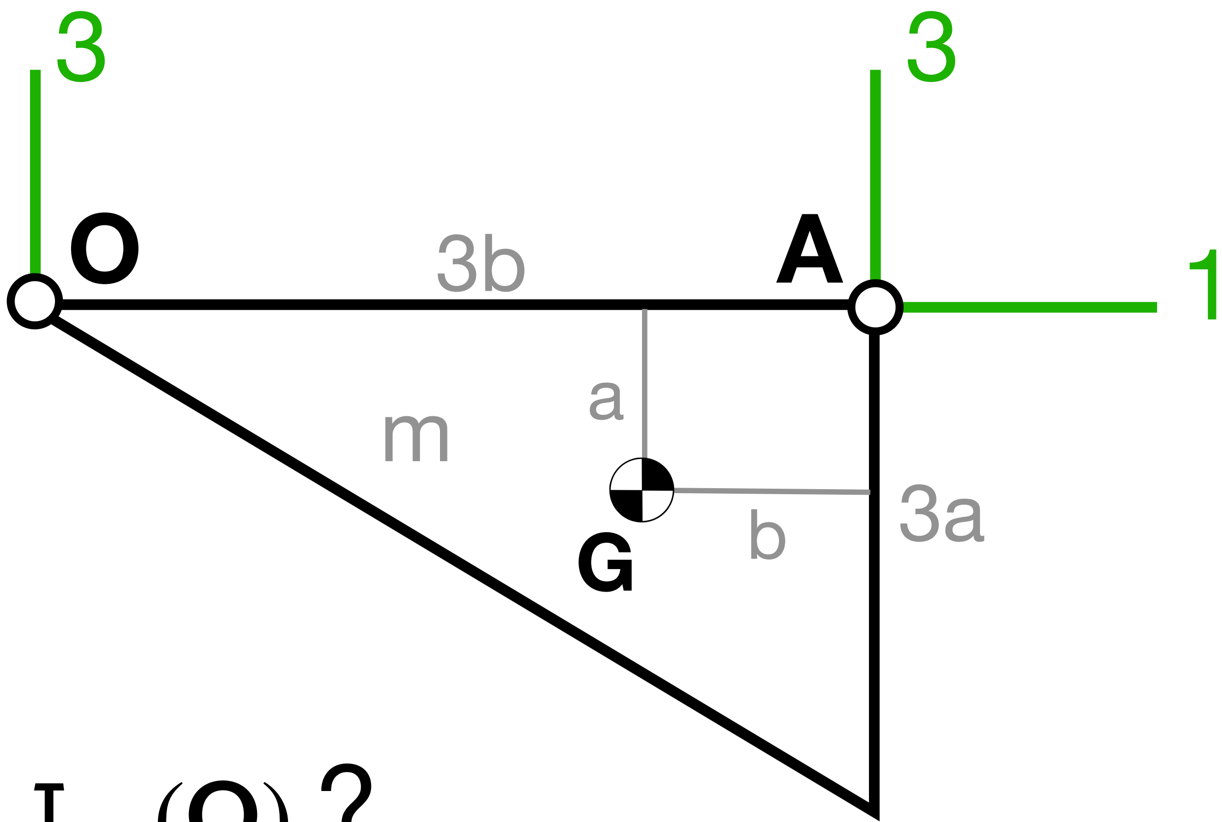
La linealitzem $\sin\varepsilon \approx \varepsilon$

$$I_{33}\,\ddot{\varepsilon} + \underbrace{9cb^2}_{\mathbf{A}}\,\dot{\varepsilon} + \underbrace{(k - 3bc\mathsf{v}_0)}_{\mathbf{B}}\,\varepsilon = 0$$

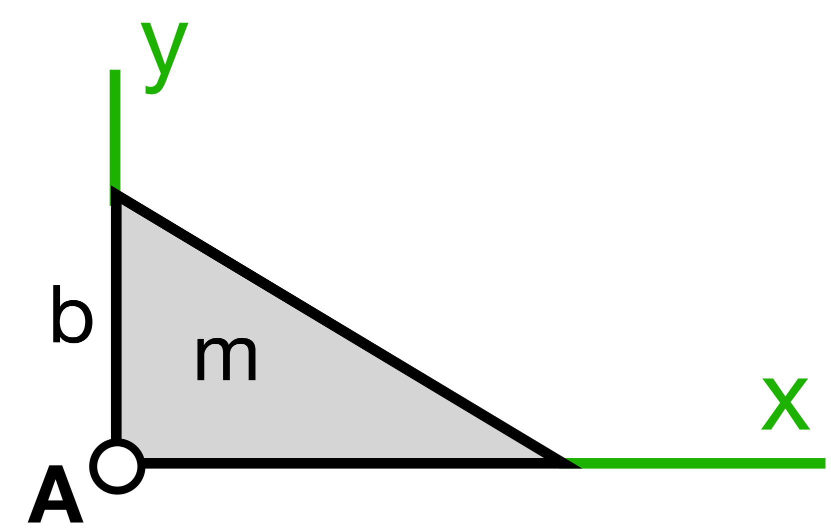
$$\ddot{\varepsilon} = -\underbrace{\frac{\mathsf{B}}{I_{33}}}_{\mathbf{K}}\,\varepsilon - \underbrace{\frac{\mathsf{A}}{I_{33}}}_{\mathbf{C} > 0}\,\dot{\varepsilon}$$

K > 0?

$$\mathbf{K} > 0 \iff \mathsf{B} > 0 \iff k > 3bc\mathsf{v}_0$$



$I_{33}(\mathbf{O})$?



$$I_{xx}(\mathbf{A}) = \frac{1}{6}mb^2$$

$I_{33}(\mathbf{A})$ de taules + **doble Steiner** per passar de **A** a **O**:

$$(a) \qquad I_{33}(\mathbf{O}) = I_{33}(\mathbf{G}) + I_{33}^{\oplus}(\mathbf{O})$$

$$(b) \qquad I_{33}(\mathbf{A}) = I_{33}(\mathbf{G}) + I_{33}^{\oplus}(\mathbf{A})$$

$$(a-b) \qquad I_{33}(\mathbf{O}) = I_{33}(\mathbf{A}) + I_{33}^{\oplus}(\mathbf{O}) - I_{33}^{\oplus}(\mathbf{A})$$

$$I_{33}(\mathbf{O}) = \frac{1}{6}m(3b)^2 + m(2b)^2 - mb^2 = \frac{9}{2}mb^2$$

DEURES

Determineu el valor de la normal N del terra sobre la placa en funció de les variables d'estat del sistema

Pistes

- Apliqueu el full de ruta

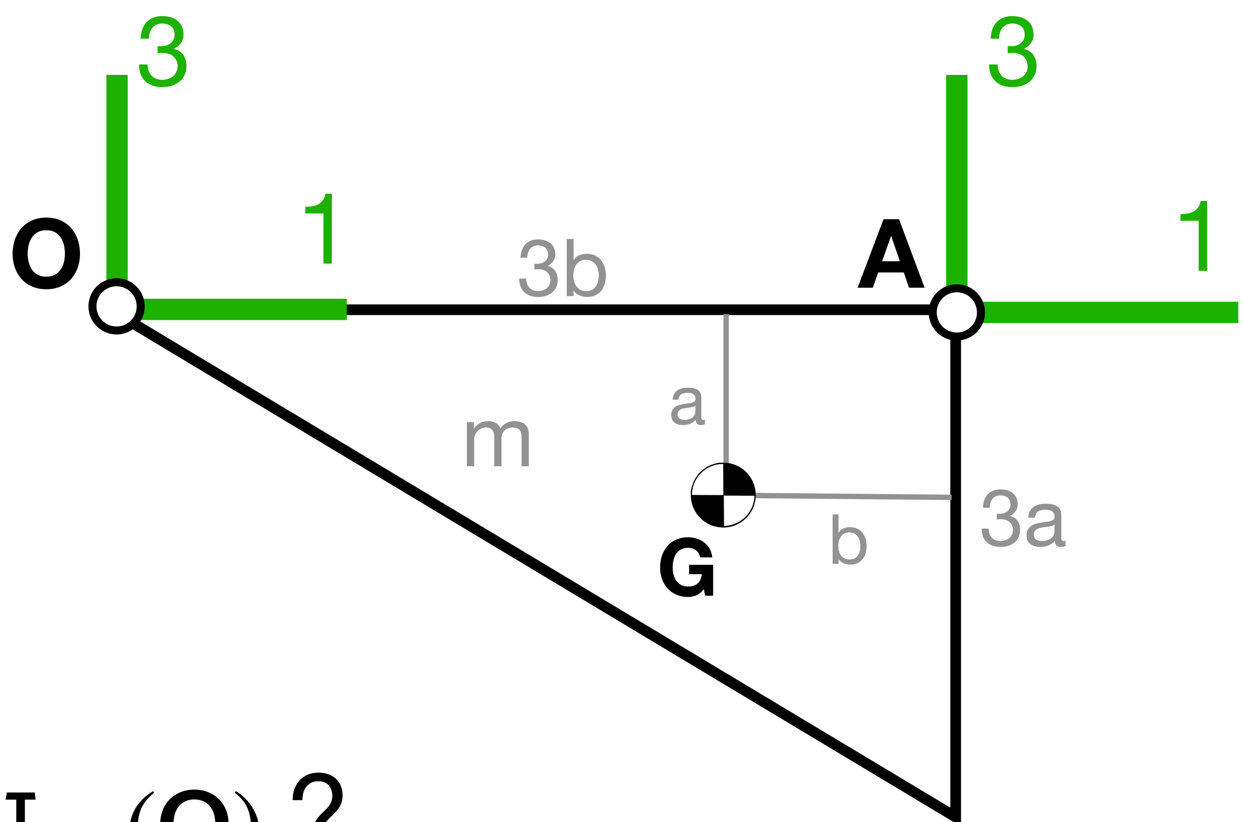
  SISTEMA = Placa

  $\text{TMC}(\mathbf{O})\Big]_2$ (aprofiteu el moment cinètic calculat abans per a l'eq. del mov.)

- Us caldrà I_{13} (vegeu-ne el càlcul a la següent transparència)

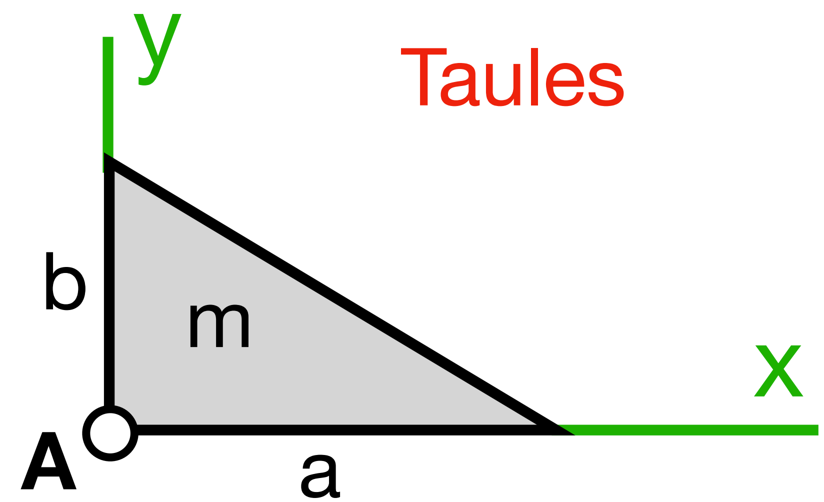
Solució

$$\mathrm{N} = \frac{2}{3}mg + \frac{ac}{b}\mathrm{v}_0\cos\theta - \frac{3}{4}ma\dot{\theta}^2$$



$I_{13}(\mathbf{O})$?

Taules



$$I_{xy}(\mathbf{A}) = -\frac{1}{12}\,m\,a\,b$$

Negatiu perquè la massa del triangle és al primer quadrant de **A** (adapteu signe al vostre cas)

$I_{13}(\mathbf{A})$ de taules + **doble Steiner** per passar de **A** a **O**:

$$I_{13}(\mathbf{O}) = I_{13}(\mathbf{A}) + I^{\oplus}_{13}(\mathbf{O}) - I^{\oplus}_{13}(\mathbf{A})$$

$$I_{13}(\mathbf{O}) = -\frac{1}{12}\,m\,(3a)\,(3b) + m\,2b\,a - \left(-m\,a\,b\right) = \frac{9}{4}\,m\,a\,b$$

Negatiu perquè tota la massa de la placa és al 3er quadrant de **A**

Positiu perquè la massa concentrada a **G** és al 4rt quadrant de **O**

Negatiu perquè la massa concentrada a **G** és al 3er quadrant de **A**

Surt positiu perquè tota la massa de la placa és al 4rt quadrant de **O**

DEURES

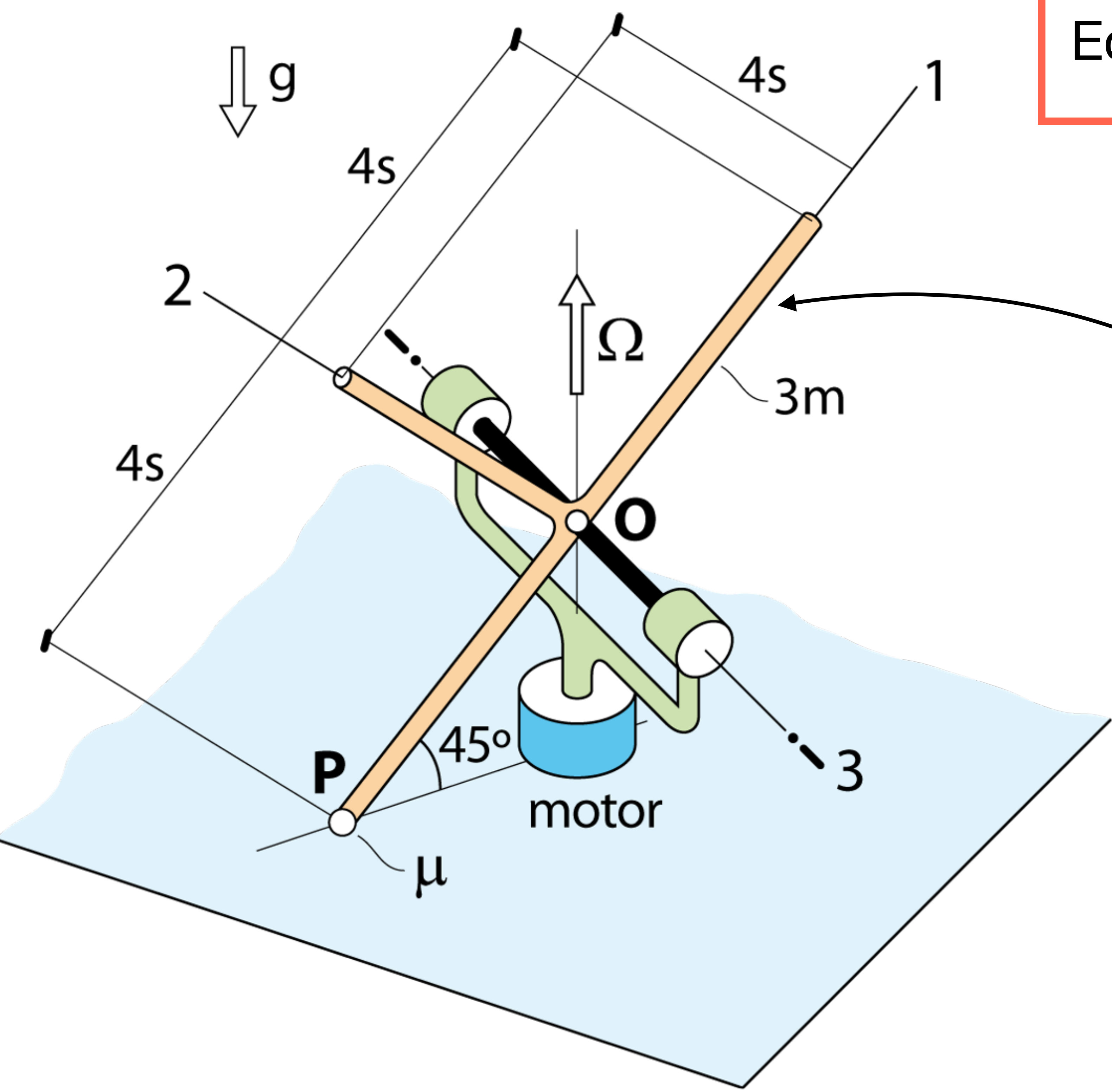
Finalment, per practicar, calculeu **TOT** el tensor d'inèrcia de la placa referit al punt **O**

$$\Omega_T^{\text{forq}} = \Omega = \text{ct}$$

Forq. de massa ≈ 0

Ω_{critica} per pèrdua contacte a P?

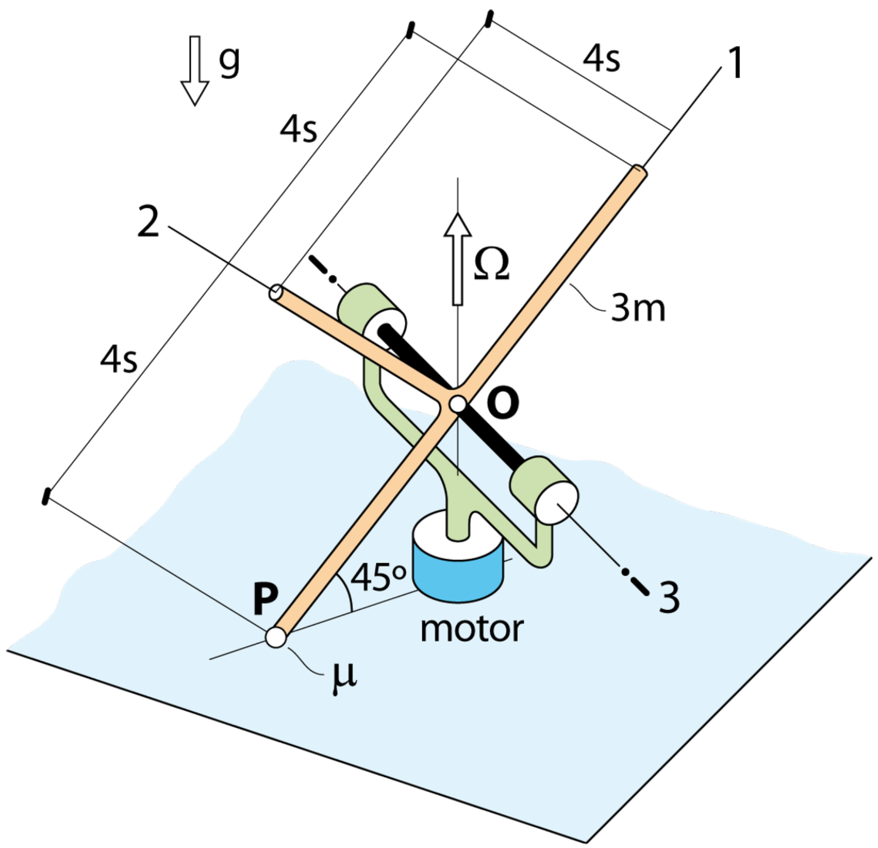
Eq. mov. quan $\Omega > \Omega_{\text{critica}}$



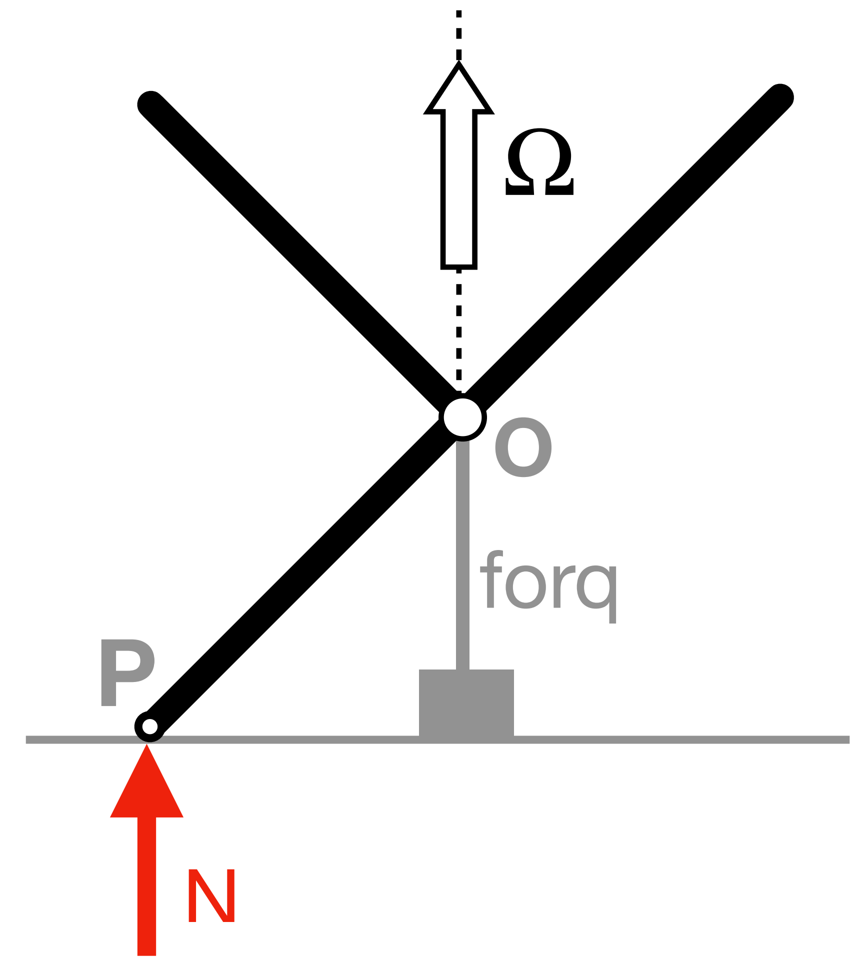
"Peça"

GL?

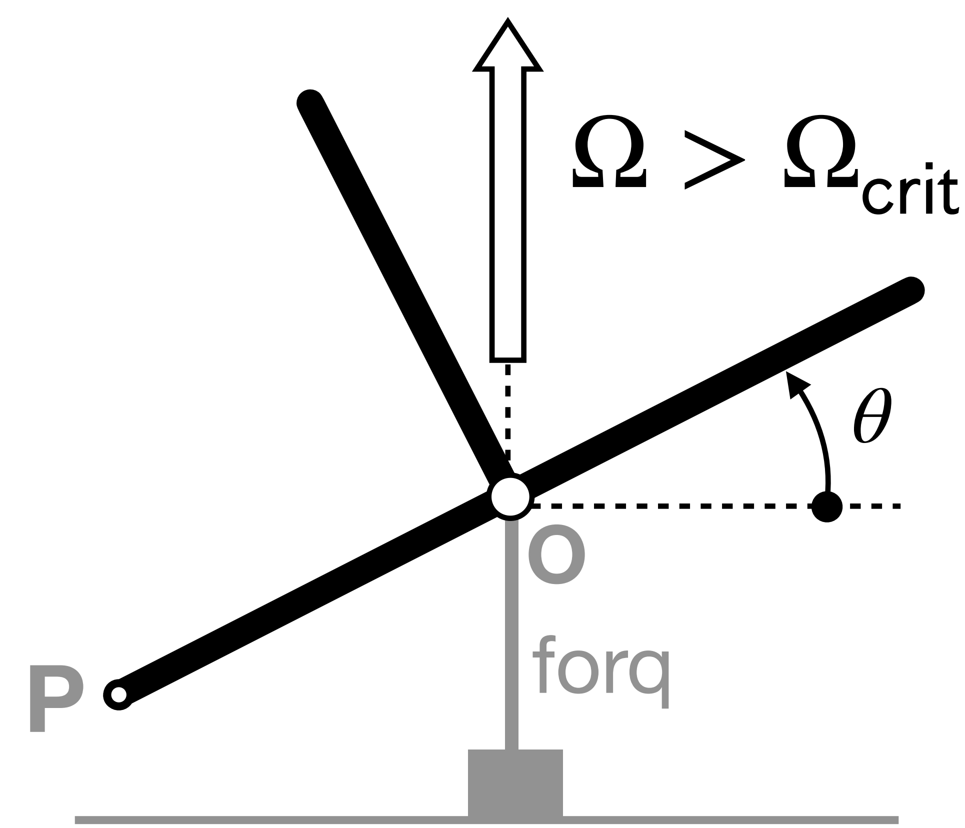
2 situacions!

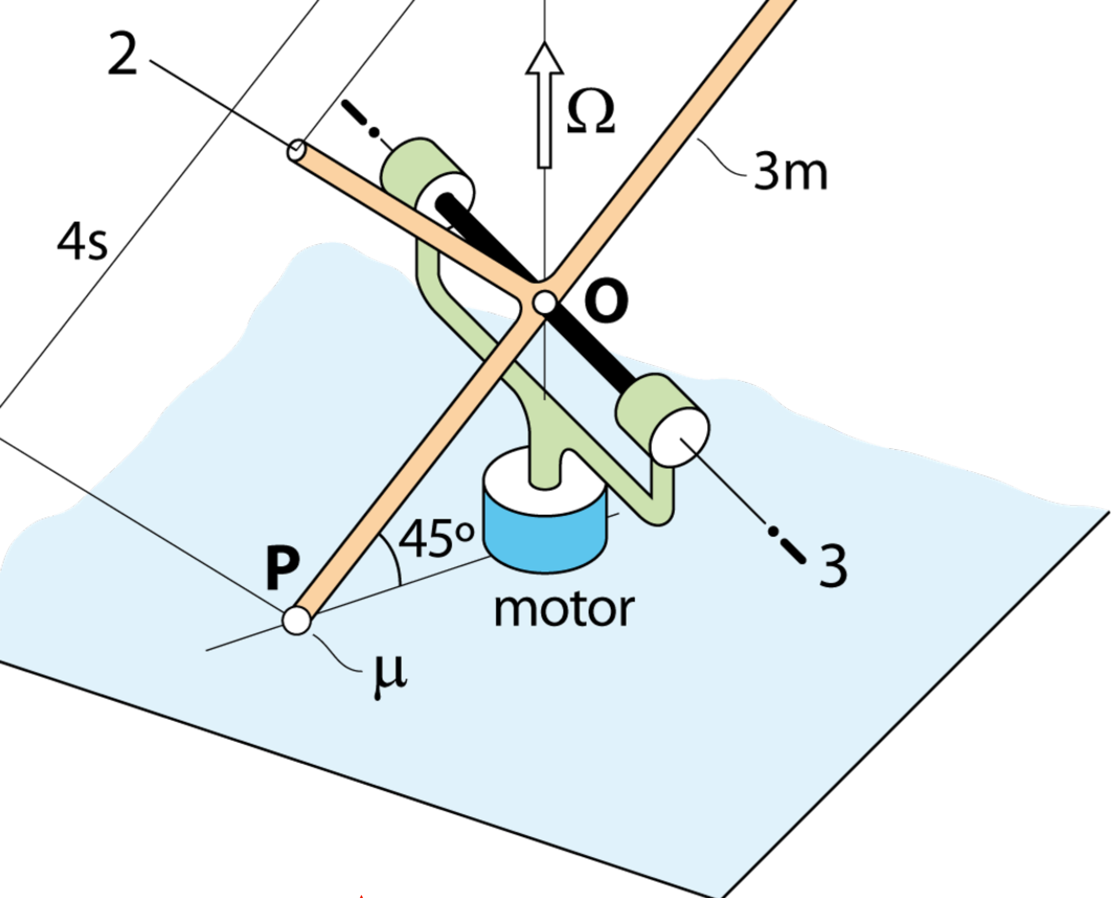


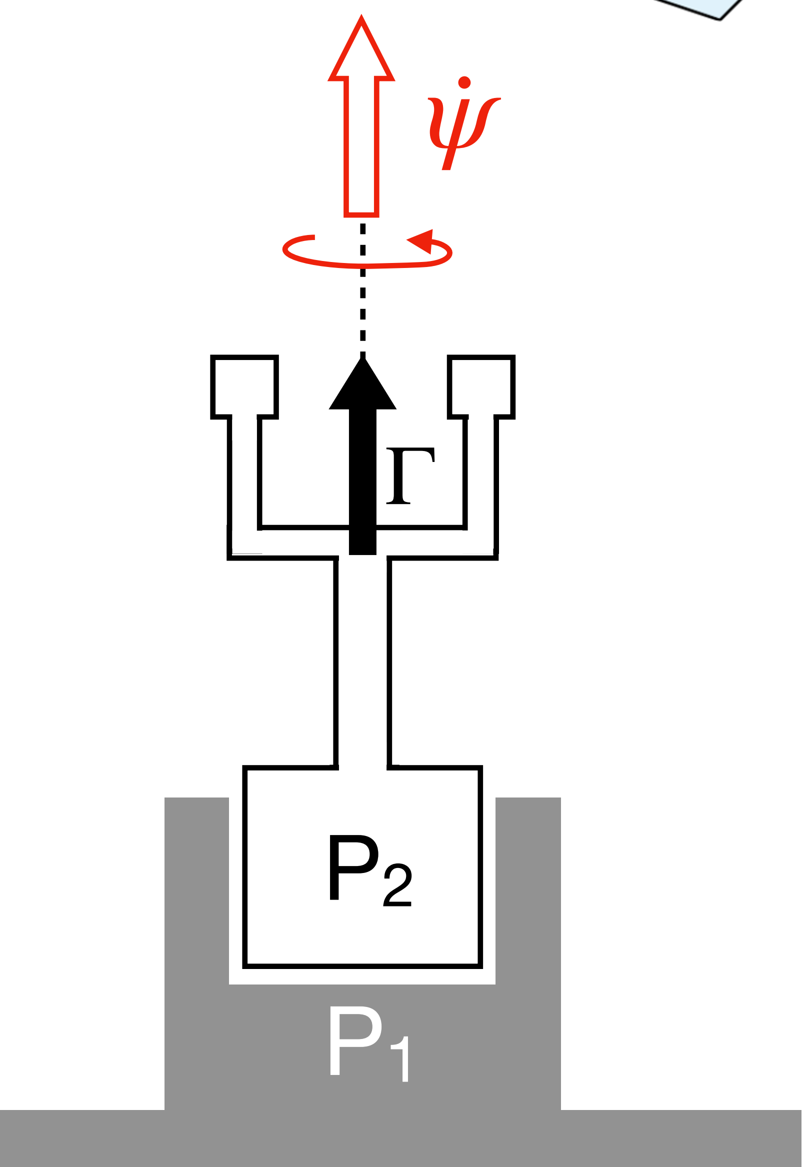
P manté contacte amb T



Contacte perdut







Recordeu

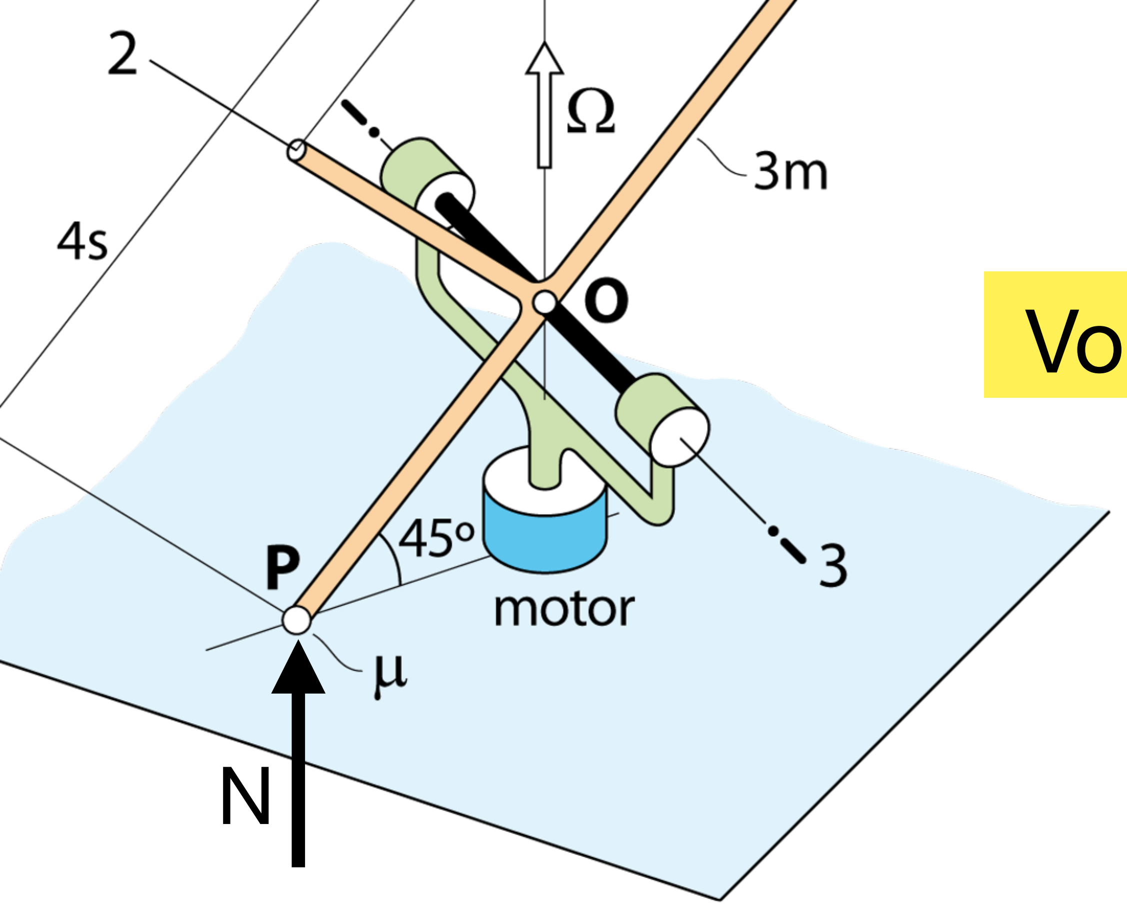
Γ conegut $\Rightarrow$ $\ddot{\psi}$ és incògnita

$\ddot{\psi}$ coneguda $\Rightarrow$ Γ és incògnita

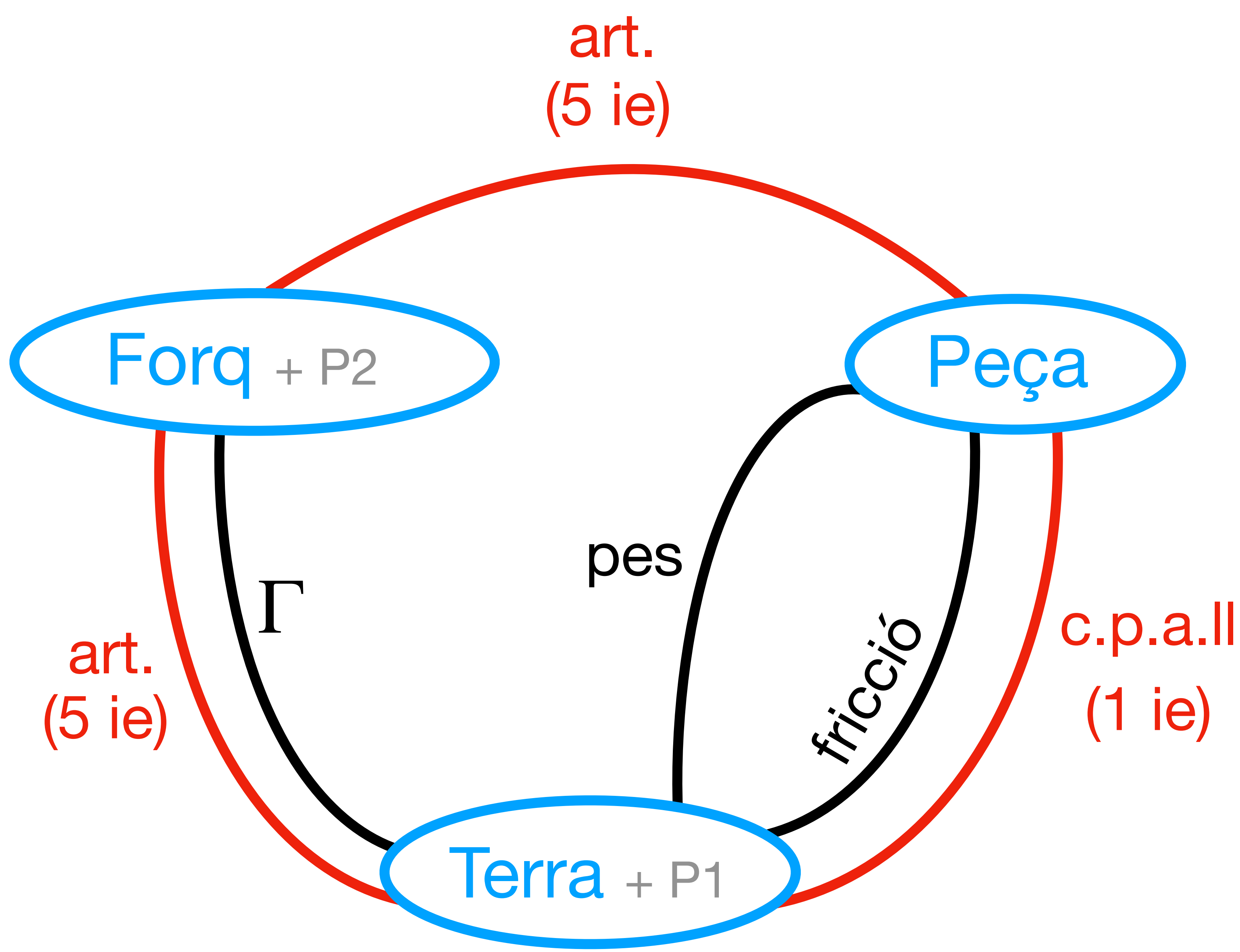
En aquest exercici

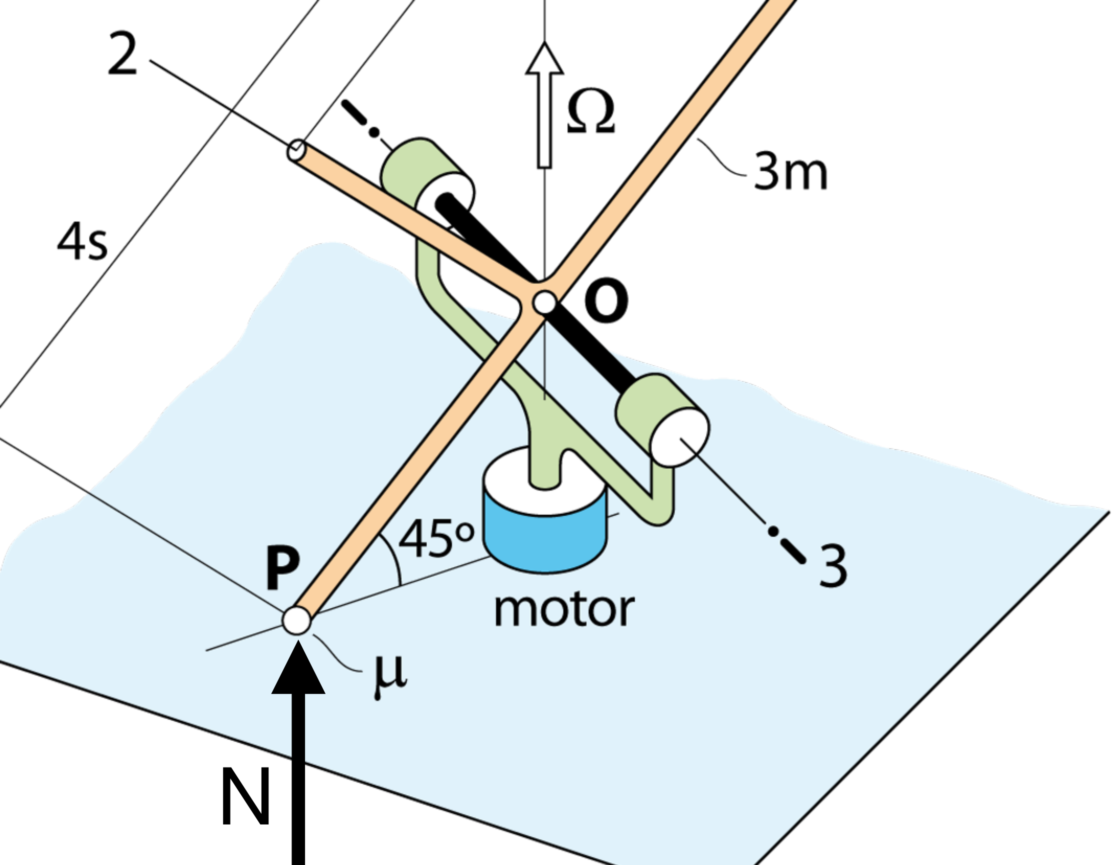
$\dot{\psi} = \Omega = ct \quad \Rightarrow \quad \ddot{\psi} = 0$ (coneguda)

Γ serà incògnita

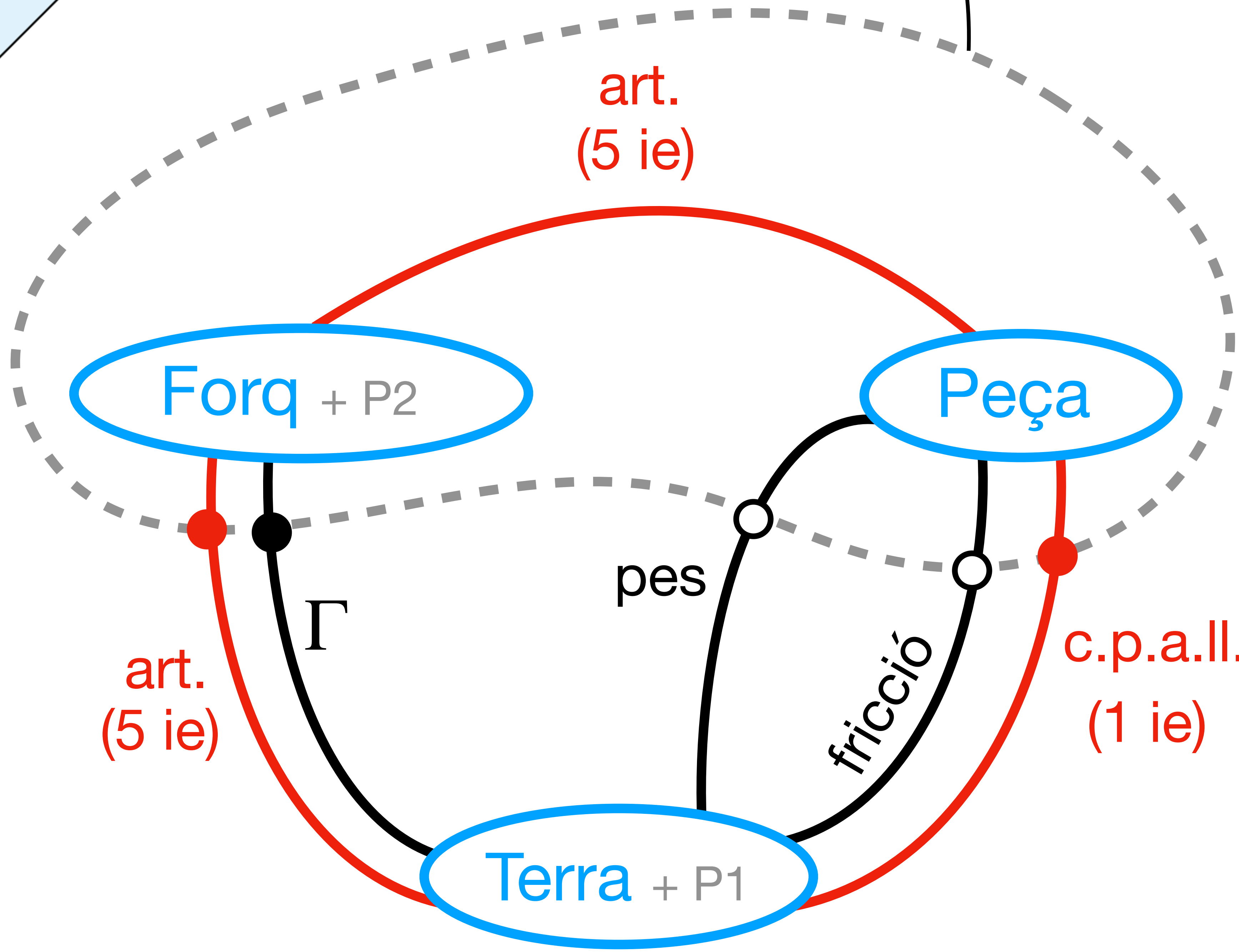


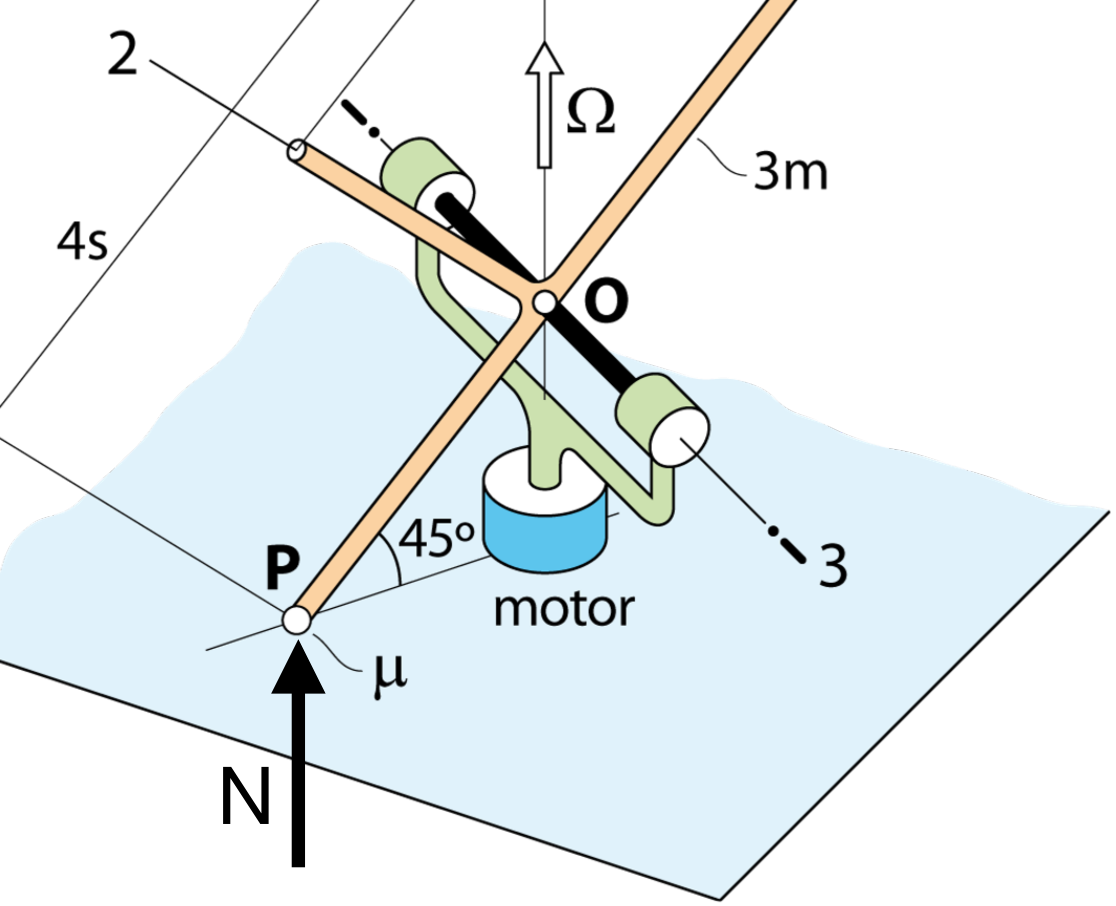
Volem N ⇒ SIST ha d'incloure la peça!

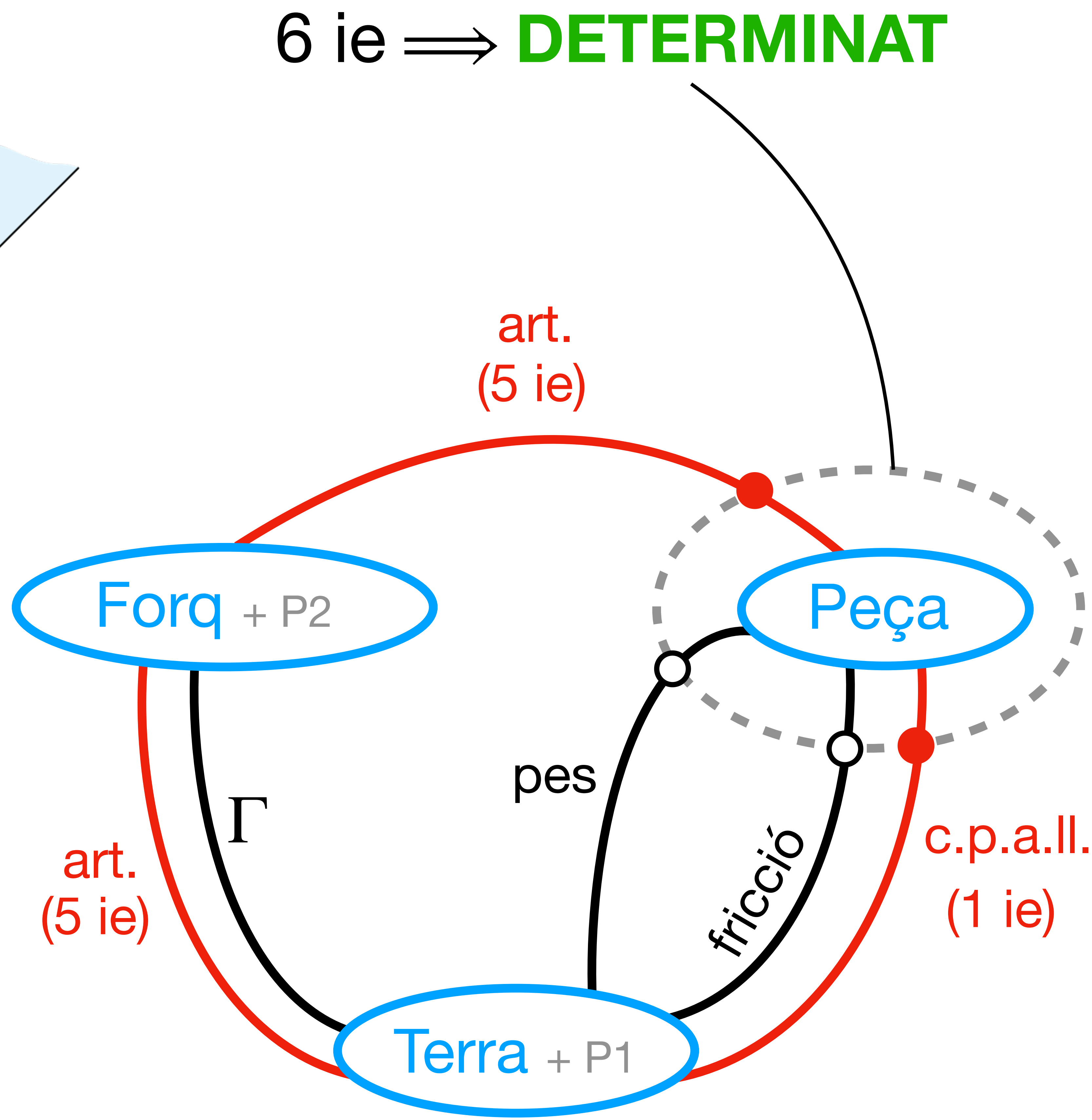


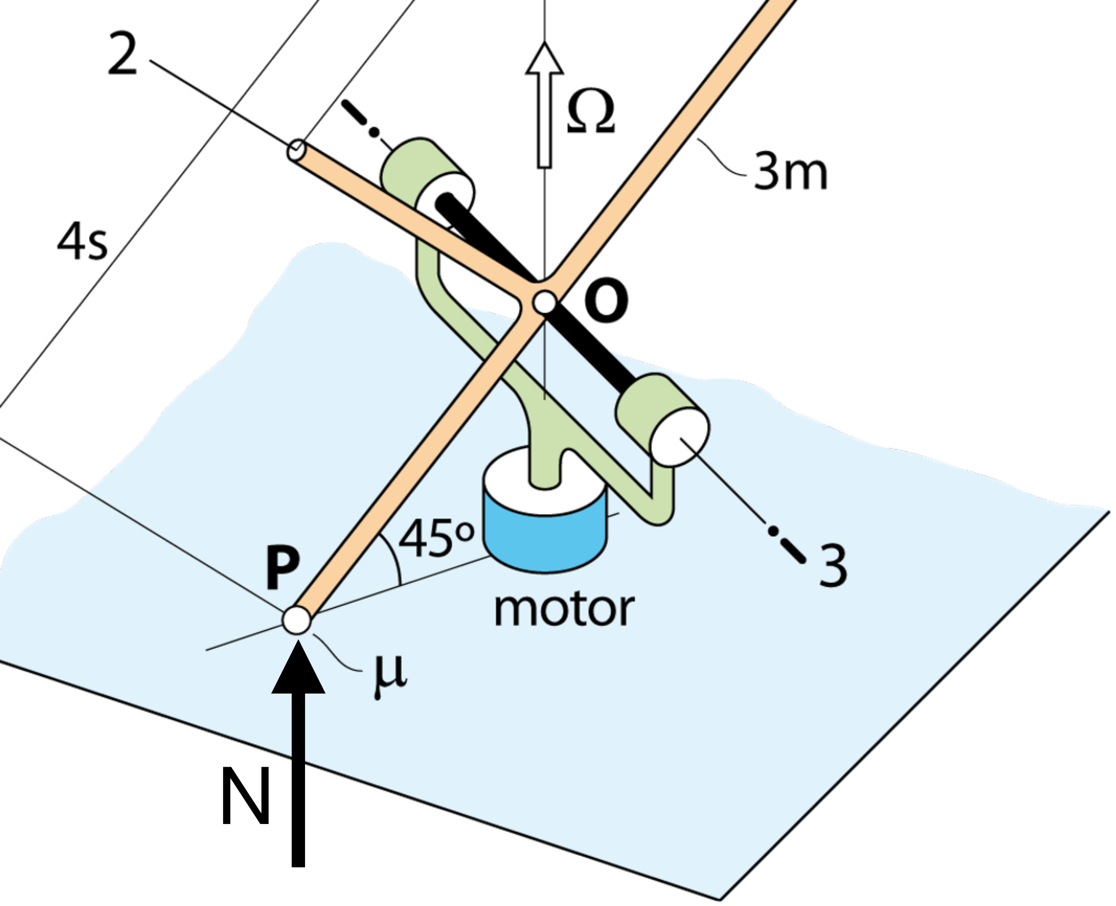


6 ie + $\Gamma \Longrightarrow$ **INDETERMINAT**

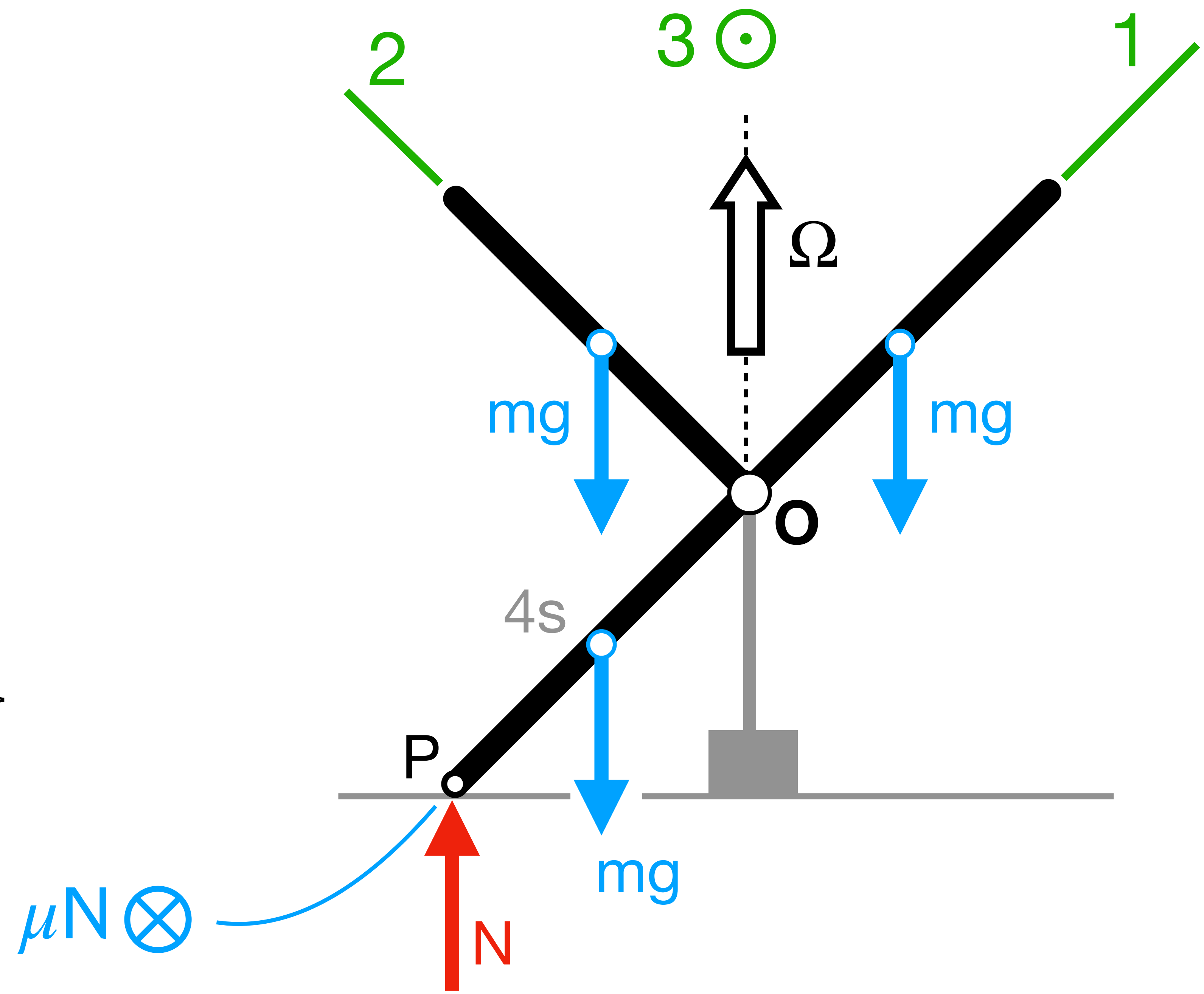



2
Ω
3m
4s
O
45º
P
motor
3
μ
N


6 ie ⟹ DETERMINAT
art.
(5 ie)
Forq + P2
Peça
art.
(5 ie)
Γ
pes
fricció
c.p.a.ll.
(1 ie)
Terra + P1



Forces sobre "Peça"



$$\left\{\bar{F}_{\text{Forq}\to\text{Peça}}\right\}_B = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

$$\left\{\bar{M}_{\text{Forq}\to\text{Peça}}(\mathbf{O})\right\}_B = \begin{Bmatrix} M_1 \\ M_2 \\ 0 \end{Bmatrix}$$

DEURES

Determineu

- Parell motor Γ per mantenir $\Omega = ct$
- Eq. mov. per al cas en que el contacte a P ja s'ha perdut ($\Omega > \Omega_{\text{critica}}$)